

✓ Variationa Encoders

1 Theory Questions

Assignment 3: Variational Autoencoders — Theory Questions

Q1: Why is the KL Divergence term important in the VAE loss function?

The Kullback–Leibler (KL) divergence term:

$$D_{\text{KL}}(q_{\phi}(z|x) \parallel p(z)) = \int q_{\phi}(z|x) \log \frac{q_{\phi}(z|x)}{p(z)} dz$$

serves as a regularization mechanism in the VAE loss function. It encourages the approximate posterior \$

$$q_{\phi}(z|x)$$

, produced by the encoder, to stay close to the prior distribution

$$p(z)$$

typically a standard normal. This ensures that:

- The latent space remains continuous and well-structured, enabling smooth interpolation and sampling.
- The model avoids overfitting by discouraging overly complex or disjoint latent representations.
- Generative sampling from the prior results in valid outputs, as the decoder is trained on latent codes drawn from a distribution close to the prior.

Without this term, the encoder might collapse into encoding inputs at arbitrary locations in the latent space, leading to disorganized and discontinuous manifolds.

References:

- Kingma & Welling, *Auto-Encoding Variational Bayes*, arXiv:1312.6114 [\[Paper\]](#)
 - Ye et al., *Distribution Matching VAE*, arXiv:2512.07778
 - Lee et al., *Latent Diffusion Models with Masked Autoencoders*, ICCV 2025
 - Stanford CS231n Lecture Notes on Generative Models
-

Q2: How does the reparameterization trick enable backpropagation through the stochastic layers of a VAE?

The reparameterization trick rewrites the stochastic sampling of the latent variable

$$z = \mu_{\phi}(x) + \sigma_{\phi}(x) \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

This reformulation isolates the stochasticity in (ϵ) , which is independent of the model parameters. The result is a differentiable computation graph with respect to (ϕ) , enabling gradient-based optimization via backpropagation. This approach offers low-variance gradient estimates and allows VAEs to be trained using standard gradient descent techniques.

References:

- Kingma & Welling, *Auto-Encoding Variational Bayes*, arXiv:1312.6114
 - Doersch, *Tutorial on Variational Autoencoders*, arXiv:1606.05908
 - Stanford CS231n Lecture Notes on Generative Models
-

Q3: Why does a VAE use a probabilistic latent space instead of a fixed latent space?

A VAE maps inputs to distributions over latent variables rather than fixed encodings. This design choice provides several advantages:

- **Uncertainty Modeling:** The encoder reflects uncertainty in its representation, which is useful for ambiguous or noisy inputs.
- **Smooth Latent Space:** Probabilistic mappings prevent the latent space from forming disconnected clusters, facilitating interpolation and sampling.
- **Generative Capability:** Sampling from the prior or learned distributions leads to plausible outputs, which wouldn't be feasible with deterministic encodings.

- **Variational Inference:** The ELBO objective requires modeling ($q_{\phi}(z|x)$) as a distribution to approximate the intractable true posterior.

In contrast, deterministic latent codes do not support sampling-based generation and often lead to poor generalization.

References:

- Doersch, *Tutorial on Variational Autoencoders*, arXiv:1606.05908
 - Ye et al., *Distribution Matching VAE*, arXiv:2512.07778
 - Lee et al., *Latent Diffusion Models with Masked Autoencoders*, ICCV 2025
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Q4: What role does KL Divergence play in ensuring a smooth latent space?

The KL divergence term encourages each latent distribution ($q_{\phi}(z|x)$) to remain close to the global prior ($p(z)$). This results in:

- **Overlap Among Latent Distributions:** Ensures that latent codes from different inputs share regions in the latent space.
- **Continuity:** Enables smooth transitions and interpolation between encodings.
- **Robust Generation:** Any point sampled from the latent space is likely to correspond to a meaningful output.
- **Decoder Robustness:** Forces the decoder to generalize across neighborhoods in latent space, not just point estimates.

By shaping the latent space into a single, coherent manifold, the KL term guarantees that the space supports generation, interpolation, and representation learning.

References:

- Kingma & Welling, *Auto-Encoding Variational Bayes*, arXiv:1312.6114
- Shrivastava et al., *Latent Space Characterization of Autoencoder Variants*, arXiv:2412.04755
- Lee et al., *Latent Diffusion Models with Masked Autoencoders*, ICCV 2025

✓ 2 Coding Task

✓ Task1

Modify the VAE architecture to use convolutional layers for both the encoder and decoder, and train it on the CIFAR-10 dataset. This modification will allow the model to capture spatial relationships within images more effectively, improving its ability to generate high-quality images. After training, compare the generated images with those from a fully connected VAE.

References / Citations

- Lightning-Universe. (n.d.). lightning-bolts [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/Lightning-Universe/lightning-bolts>
- Lightning-Universe. (n.d.). basic_vae_module.py (lightning-bolts) [Source code]. GitHub. Retrieved January 31, 2026, from https://github.com/Lightning-Universe/lightning-bolts/blob/master/src/pl_bolts/models/autoencoders/basic_vae/basic_vae_module.py
- pi-tau. (n.d.). vae [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/pi-tau/vae>
- SashaMalysheva. (n.d.). Pytorch-VAE [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/SashaMalysheva/Pytorch-VAE>

```
1 ##### Cell 1.1 Imports + Reproducibility
2 import os, math, random
3 import numpy as np
4 import matplotlib.pyplot as plt
5
6 import torch
7 import torch.nn as nn
8 import torch.nn.functional as F
9 from torch.utils.data import DataLoader
10
11 import torchvision
12 import torchvision.transforms as T
13 from torchvision.utils import make_grid
14
15 from tqdm.auto import tqdm
16
17
18 def seed_everything(seed=42):
```

```
19 random.seed(seed)
20 np.random.seed(seed)
21 torch.manual_seed(seed)
22 torch.cuda.manual_seed_all(seed)
23 torch.backends.cudnn.deterministic = False
24 torch.backends.cudnn.benchmark = True
25
26 seed_everything(42)
27
28 device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
29 print("Device:", device)
30
```

Device: cpu

```
1 ### Cell 1.2 - CIFAR-10 DataLoaders
2 BATCH_SIZE = 128
3 NUM_WORKERS = 2 # can be increase
4
5 transform = T.Compose([
6     T.ToTensor(), # outputs in [0,1]
7 ])
8
9 train_ds = torchvision.datasets.CIFAR10(root="./data", train=True, download=True, transform=transform)
10 test_ds = torchvision.datasets.CIFAR10(root="./data", train=False, download=True, transform=transform)
11
12 train_loader = DataLoader(train_ds, batch_size=BATCH_SIZE, shuffle=True,
13                             num_workers=NUM_WORKERS, pin_memory=(device.type=="cuda"))
14 test_loader = DataLoader(test_ds, batch_size=BATCH_SIZE, shuffle=False,
15                             num_workers=NUM_WORKERS, pin_memory=(device.type=="cuda"))
16
17 print("Train batches:", len(train_loader), "| Test batches:", len(test_loader))
18
```

100%|██████████| 170M/170M [00:02<00:00, 78.7MB/s]
Train batches: 391 | Test batches: 79

```

1 ### Cell 1.3 – Helper Functions (Loss, Sampling, Plotting)
2 def kl_divergence(mu, logvar):
3     # KL(q(z|x) || p(z)) where p(z)=N(0,1)
4     # returns sum over batch and latent dims
5     return -0.5 * torch.sum(1 + logvar - mu.pow(2) - logvar.exp())
6
7 def recon_loss_fn(recon_x, x, kind="mse"):
8     # returns sum over all pixels + batch
9     if kind == "bce":
10         return F.binary_cross_entropy(recon_x, x, reduction="sum")
11     return F.mse_loss(recon_x, x, reduction="sum")
12
13 def vae_loss(recon_x, x, mu, logvar, beta=1.0, recon_kind="mse"):
14     recon = recon_loss_fn(recon_x, x, kind=recon_kind)
15     kl = kl_divergence(mu, logvar)
16     total = recon + beta * kl
17     # Return per-batch average (more readable)
18     bs = x.size(0)
19     return total / bs, recon / bs, kl / bs
20
21 @torch.no_grad()
22 def sample_images(model, latent_dim, n=8):
23     model.eval()
24     z = torch.randn(n, latent_dim, device=device)
25     x = model.decode(z).clamp(0, 1)
26     return x
27
28 @torch.no_grad()
29 def reconstruct_images(model, x):
30     model.eval()
31     recon, mu, logvar = model(x)
32     return recon.clamp(0, 1)
33
34 def show_grid(images, title="", nrow=8, figsize=(10, 4)):
35     # images: (B,3,32,32)
36     grid = make_grid(images.cpu(), nrow=nrow, padding=2)
37     plt.figure(figsize=figsize)
38     plt.imshow(grid.permute(1,2,0))
39     plt.axis("off")

```

```
40 plt.title(title)
41 plt.show()
42
```

```
1 ### Cell 1.4 Cell 4 – Baseline: Fully-Connected VAE (MLP)
2 class FCVAE(nn.Module):
3     def __init__(self, latent_dim=128, hidden_dims=(1024, 512)):
4         super().__init__()
5         self.latent_dim = latent_dim
6         in_dim = 3 * 32 * 32
7
8         h1, h2 = hidden_dims
9         self.enc = nn.Sequential(
10             nn.Linear(in_dim, h1),
11             nn.ReLU(inplace=True),
12             nn.Linear(h1, h2),
13             nn.ReLU(inplace=True),
14         )
15         self.fc_mu = nn.Linear(h2, latent_dim)
16         self.fc_logvar = nn.Linear(h2, latent_dim)
17
18         self.dec_fc = nn.Sequential(
19             nn.Linear(latent_dim, h2),
20             nn.ReLU(inplace=True),
21             nn.Linear(h2, h1),
22             nn.ReLU(inplace=True),
23             nn.Linear(h1, in_dim),
24         )
25
26     def encode(self, x):
27         x = x.view(x.size(0), -1)
28         h = self.enc(x)
29         return self.fc_mu(h), self.fc_logvar(h)
30
31     def reparameterize(self, mu, logvar):
32         std = torch.exp(0.5 * logvar)
33         eps = torch.randn_like(std)
34         return mu + eps * std
```

```

35
36     def decode(self, z):
37         out = self.dec_fc(z)
38         out = torch.sigmoid(out) # output in [0,1]
39         return out.view(z.size(0), 3, 32, 32)
40
41     def forward(self, x):
42         mu, logvar = self.encode(x)
43         z = self.reparameterize(mu, logvar)
44         recon = self.decode(z)
45         return recon, mu, logvar
46

```

```

1 ### Cell 1.5 – Task Model: Convolutional VAE (Conv Encoder + Conv Decoder)
2 class ConvVAE(nn.Module):
3     def __init__(self, latent_dim=128):
4         super().__init__()
5         self.latent_dim = latent_dim
6
7         # 32x32 -> 16x16 -> 8x8 -> 4x4 -> 2x2
8         self.encoder = nn.Sequential(
9             nn.Conv2d(3, 32, 4, 2, 1), # (B,32,16,16)
10            nn.BatchNorm2d(32),
11            nn.ReLU(inplace=True),
12
13            nn.Conv2d(32, 64, 4, 2, 1), # (B,64,8,8)
14            nn.BatchNorm2d(64),
15            nn.ReLU(inplace=True),
16
17            nn.Conv2d(64, 128, 4, 2, 1), # (B,128,4,4)
18            nn.BatchNorm2d(128),
19            nn.ReLU(inplace=True),
20
21            nn.Conv2d(128, 256, 4, 2, 1), # (B,256,2,2)
22            nn.BatchNorm2d(256),
23            nn.ReLU(inplace=True),
24        )
25

```



```

26     self.enc_out_dim = 256 * 2 * 2
27     self.fc_mu = nn.Linear(self.enc_out_dim, latent_dim)
28     self.fc_logvar = nn.Linear(self.enc_out_dim, latent_dim)
29
30     self.fc_dec = nn.Linear(latent_dim, self.enc_out_dim)
31
32     # 2x2 -> 4x4 -> 8x8 -> 16x16 -> 32x32
33     self.decoder = nn.Sequential(
34         nn.ConvTranspose2d(256, 128, 4, 2, 1), # (B,128,4,4)
35         nn.BatchNorm2d(128),
36         nn.ReLU(inplace=True),
37
38         nn.ConvTranspose2d(128, 64, 4, 2, 1), # (B,64,8,8)
39         nn.BatchNorm2d(64),
40         nn.ReLU(inplace=True),
41
42         nn.ConvTranspose2d(64, 32, 4, 2, 1), # (B,32,16,16)
43         nn.BatchNorm2d(32),
44         nn.ReLU(inplace=True),
45
46         nn.ConvTranspose2d(32, 3, 4, 2, 1), # (B,3,32,32)
47         nn.Sigmoid(), # output in [0,1]
48     )
49
50     def encode(self, x):
51         h = self.encoder(x)
52         h = h.view(h.size(0), -1)
53         return self.fc_mu(h), self.fc_logvar(h)
54
55     def reparameterize(self, mu, logvar):
56         std = torch.exp(0.5 * logvar)
57         eps = torch.randn_like(std)
58         return mu + eps * std
59
60     def decode(self, z):
61         h = self.fc_dec(z).view(z.size(0), 256, 2, 2)
62         return self.decoder(h)
63
64     def forward(self, x):

```

```
65     mu, logvar = self.encode(x)
66     z = self.reparameterize(mu, logvar)
67     recon = self.decode(z)
68     return recon, mu, logvar
69
```

```
1 ### Cel 1.6 – Training / Evaluation Loop
2 def train_vae(model, train_loader, test_loader, epochs=10, lr=2e-4, beta=1.0, recon_kind="mse"):
3     model = model.to(device)
4     opt = torch.optim.Adam(model.parameters(), lr=lr)
5
6     use_amp = (device.type == "cuda")
7     scaler = torch.cuda.amp.GradScaler(enabled=use_amp)
8
9     history = {"train_total": [], "train_recon": [], "train_kl": [],
10               "test_total": [], "test_recon": [], "test_kl": []}
11
12     for epoch in range(1, epochs + 1):
13         # ---- train ----
14         model.train()
15         t_total = t_recon = t_kl = 0.0
16         n_train = 0
17
18         pbar = tqdm(train_loader, desc=f"Epoch {epoch}/{epochs} [train]", leave=False)
19         for x, _ in pbar:
20             x = x.to(device, non_blocking=True)
21
22             opt.zero_grad(set_to_none=True)
23             with torch.cuda.amp.autocast(enabled=use_amp):
24                 recon, mu, logvar = model(x)
25                 loss, rloss, kl = vae_loss(recon, x, mu, logvar, beta=beta, recon_kind=recon_kind)
26
27             scaler.scale(loss).backward()
28             scaler.step(opt)
29             scaler.update()
30
31             bs = x.size(0)
32             t_total += loss.item() * bs
```

```

33         t_recon += rloss.item() * bs
34         t_kl += kl.item() * bs
35         n_train += bs
36
37         pbar.set_postfix({"loss": loss.item(), "recon": rloss.item(), "kl": kl.item()})
38
39     train_total = t_total / n_train
40     train_recon = t_recon / n_train
41     train_kl = t_kl / n_train
42
43     # ---- eval ----
44     model.eval()
45     v_total = v_recon = v_kl = 0.0
46     n_test = 0
47     with torch.no_grad():
48         for x, _ in tqdm(test_loader, desc=f"Epoch {epoch}/{epochs} [eval]", leave=False):
49             x = x.to(device, non_blocking=True)
50             recon, mu, logvar = model(x)
51             loss, rloss, kl = vae_loss(recon, x, mu, logvar, beta=beta, recon_kind=recon_kind)
52
53             bs = x.size(0)
54             v_total += loss.item() * bs
55             v_recon += rloss.item() * bs
56             v_kl += kl.item() * bs
57             n_test += bs
58
59     test_total = v_total / n_test
60     test_recon = v_recon / n_test
61     test_kl = v_kl / n_test
62
63     history["train_total"].append(train_total)
64     history["train_recon"].append(train_recon)
65     history["train_kl"].append(train_kl)
66     history["test_total"].append(test_total)
67     history["test_recon"].append(test_recon)
68     history["test_kl"].append(test_kl)
69
70     print(f"Epoch {epoch:02d}/{epochs} | "
71           f"Train: total={train_total:.4f}, recon={train_recon:.4f}, kl={train_kl:.4f} | ")

```

```
72         f"Test: total={test_total:.4f}, recon={test_recon:.4f}, kl={test_kl:.4f}")
73
74     return history
75
```

```
1 ##### Cell 1.7 – Train FC-VAE and Conv-VAE
2 LATENT_DIM = 128
3 EPOCHS = 50          # try 5 for a quick run, 20-50 for better samples
4 LR = 2e-4
5 BETA = 1.5           # classic VAE; trying 1 - 0.5 or 2.0 v=cane be tweaked to see effect
6 RECON = "mse"        # "mse" or "bce" (MSE tends to behave better for natural images)
7
8 fc_vae = FCVAE(latent_dim=LATENT_DIM)
9 conv_vae = ConvVAE(latent_dim=LATENT_DIM)
10
11 print("Training Fully-Connected VAE...")
12 hist_fc = train_vae(fc_vae, train_loader, test_loader,
13                     epochs=EPOCHS, lr=LR, beta=BETA, recon_kind=RECON)
14
15 print("\nTraining Convolutional VAE...")
16 hist_conv = train_vae(conv_vae, train_loader, test_loader,
17                       epochs=EPOCHS, lr=LR, beta=BETA, recon_kind=RECON)
18
```



```
Training Fully-Connected VAE...  
/tmp/ipython-input-1189186631.py:7: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler(  
    scaler = torch.cuda.amp.GradScaler(enabled=use_amp)  
  
/tmp/ipython-input-1189186631.py:23: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cu  
with torch.cuda.amp.autocast(enabled=use_amp):  
  
Epoch 01/50 | Train: total=154.0918, recon=145.6712, kl=5.6137 | Test: total=135.5942, recon=122.3681, kl=8.8174  
  
Epoch 02/50 | Train: total=130.7314, recon=118.4788, kl=8.1684 | Test: total=123.2742, recon=111.1305, kl=8.0958  
  
Epoch 03/50 | Train: total=118.6342, recon=106.8392, kl=7.8633 | Test: total=114.9866, recon=103.5607, kl=7.6173  
  
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>  
Exception ignored in: <function _MultiProcessin  
Traceback (most recent call last):  
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del_  
Traceback (most recent call last):  
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del_  
self._shutdown_workers()  
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers  
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers  
if w.is_alive():  
^^^^^^^^^^^^^^^^^^^^^^^^^^  
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive  
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive  
assert self._parent_pid == os.getpid(), 'can only test a child process'  
assert self._parent_pid == os.getpid(), 'can only test a child process'  
^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^  
AssertionError: : can only test a child process
```

```
Exception ignored in: Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0><function _MultiProcessin
```

```
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self._shutdown_workers() File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_wor  
  
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers  
        if w.is_alive():  
if w.is_alive():  
    ^^^^^^^^^^^^^^^^^^^^^^^^  
^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive  
  
    assert self._parent_pid == os.getpid(), 'can only test a child process' File "/usr/lib/python3.12/multiprocessing/process.py", line  
  
    assert self._parent_pid == os.getpid(), 'can only test a child process'  
       ^  ^^ ^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^  
^^AssertionError^^:  
can only test a child processAssertionError  
: Exception ignored in: can only test a child process<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>  
  
Traceback (most recent call last):  
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__  
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>  
self._shutdown_workers()Traceback (most recent call last):  
  
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers  
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__  
    if w.is_alive():self._shutdown_workers()  
  
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers  
        if w.is_alive():  
^   ^   ^   ^   ^   ^   ^^^^^^^^^^^^^^^  
^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive  
^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'^  
^   ^  
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive  
        assert self._parent_pid == os.getpid(), 'can only test a child process'  
           ^  ^^ ^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^  
^AssertionError^^: ^can only test a child process^
```

[illegible]

Epoch 08/50 | Train: total=100.0688, recon=85.6052, kl=9.6424 | Test: total=99.2660, recon=84.3624, kl=9.9357

Epoch 09/50 | Train: total=98.7652, recon=83.8260, kl=9.9595 | Test: total=97.9713, recon=83.1115, kl=9.9065

```
Exception ignored in: Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>Traceback (most recent call last):
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
Traceback (most recent call last):
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()self._shutdown_workers()
```

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

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File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():if w.is_alive():
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File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

```
assert self._parent_pid == os.getpid(), 'can only test a child process'assert self._parent_pid == os.getpid(), 'can only test a
```

```
^AssertionError^: ^can only test a child process^
```

Exception ignored in:

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>AssertionError: can only test a child process
```

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

```
Exception ignored in: self._shutdown_workers() at function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0
```

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    if w.is_alive():self._shutdown_workers()
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^^^^^^^^^^^^^^^^^^^^
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
       ^ ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^AssertionError: can only test a child process

AssertionError: can only test a child processException ignored in:
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>Exception ignored in:
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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
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    if w.is_alive():if w.is_alive():

       ^^ ^^ ^^^^^^^^^^^^^^^^^^^^
^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    ^assert self._parent_pid == os.getpid(), 'can only test a child process'^

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    assert self._parent_pid == os.getpid(), 'can only test a child process'
       ^ ^ ^ ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^AssertionError: can only test a child process

Exception ignored in: AssertionError<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>:
Traceback (most recent call last):
can only test a child process File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
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```

```

self._shutdown_workers()
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
    ^   ^   ^   ^^^^^^^^^^^^^^^^^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^^^
assert self._parent_pid == os.getpid(), 'can only test a child process' File "/usr/lib/python3.12/multiprocessing/process.py", line 160
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        ^   ^   ^   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
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    self._shutdown_workers()
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
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    ^   ^   ^   ^   ^^^^^^^^^
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^   ^   ^   ^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
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    ^   ^   ^   ^   ^   ^   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
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AssertionError<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>:
can only test a child process
Exception ignored in: Traceback (most recent call last):
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

```

```

self._shutdown_workers()
    if w.is_alive(): File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
        if w.is_alive():
        ^ ^ ^ ~~~~~
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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    : can only test a child processself._shutdown_workers()

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    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
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    self._shutdown_workers()^^
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    if w.is_alive():

    AssertionError : can only test a child process
Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^

```


Epoch 11/50 | Train: total=96.1791, recon=80.9654, kl=10.1425 | Test: total=95.3884, recon=80.2953, kl=10.0621

Epoch 12/50 | Train: total=94.5442, recon=78.7315, kl=10.5418 | Test: total=94.8838, recon=79.1841, kl=10.4665

Epoch 13/50 | Train: total=93.7550, recon=77.6244, kl=10.7538 | Test: total=93.6598, recon=77.1886, kl=10.9808

Epoch 14/50 | Train: total=92.9184, recon=76.6583, kl=10.8401 | Test: total=92.7005, recon=76.1420, kl=11.0390

Epoch 15/50 | Train: total=92.2598, recon=75.9064, kl=10.9023 | Test: total=92.1085, recon=75.6106, kl=10.9986

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
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^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^if w.is_alive():^    ^if w.is_alive():
^ ^ ^ ^ ^ ^
```



```

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^ ^ ^ ^ assert self._parent_pid == os.getpid(), 'can only test a child process'^ ^
^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^
^AssertionError^: ^can only test a child process
^^
AssertionError: Exception ignored in: can only test a child process^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^

^Traceback (most recent call last):
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^   ^Exception ignored in: self._shutdown_workers()^
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>   ^^
if w.is_alive():Traceback (most recent call last):

^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^   self._shutdown_workers()^
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^   ^if w.is_alive():
^ ^
^ AssertionError^: ^can only test a child process ^^
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^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^
^   ^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
self._shutdown_workers()^
^assert self._parent_pid == os.getpid(), 'can only test a child process' File "/usr/local/lib/python3.12/dist-packages/torch/utils/data

^   ^ if w.is_alive(): ^
^ ^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'
^^ ^^ ^^ ^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^
^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'^^
^^ ^^ ^^ ^^ ^^ ^ ^ ^^ ^^ ^ ^^ ^^ ^^ ^^ ^^
^^^^AssertionError^^: ^^^can only test a child process
^^^^Exception ignored in: ^^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
^Traceback (most recent call last):
^^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^^^^   ^
^self._shutdown_workers()AssertionError^
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

```

```

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
: ^can only test a child process^
    ^if w.is_alive():^Exception ignored in:
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Traceback (most recent call last):
^ ^ ^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
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^^
^^^
^   AssertionError^if w.is_alive():^: ^can only test a child process
^^
^^
Exception ignored in:   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>assert self._parent_pid == os.getpid(), 'can only test a child

Traceback (most recent call last):
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
        self._shutdown_workers()^
^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^ ^     ^if w.is_alive(): ^
^ ^         ^ ^ ^ ^ ^ ^ ^ ^
^^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^       ^^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^^^
^ ^^^ ^ ^ ^ ^ ^ ^ ^
^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^       assert self._parent_pid == os.getpid(), 'can only test a child process'^
^^   ^ ^   ^ ^ ^ ^ ^ ^ ^ ^ ^ ^
^   AssertionError^^: ^can only test a child process^^
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^^^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^^^       ^self._shutdown_workers()^
^^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^       ^if w.is_alive():^^^^
^^ ^^^ ^ ^ ^ ^ ^ ^
^   ^AssertionError : ^can only test a child process
^^^^^Exception ignored in: ^^^^^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^^^

^Traceback (most recent call last):
AssertionError^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^       : ^self._shutdown_workers()

can only test a child process   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

```



```

    ^ ^  ^^ ^^^^^^ ^^^^^^^^^^^^^^^^^^^^^
^^  File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^    ^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^

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      ^ assert self._parent_pid == os.getpid(), 'can only test a child process'^
^  ^    ^^ ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^
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^^^^Traceback (most recent call last):
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^^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^    ^^if w.is_alive():^^^
^^  ^^^^
^AssertionError ^: ^ can only test a child process

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^^^^^^^^^
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
      assert self._parent_pid == os.getpid(), 'can only test a child process'
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
AssertionError: can only test a child process

```

Epoch 16/50 | Train: total=91.7693, recon=75.2973, kl=10.9813 | Test: total=91.7270, recon=75.1847, kl=11.0282

Epoch 17/50 | Train: total=91.2984, recon=74.7435, kl=11.0366 | Test: total=91.4638, recon=74.1529, kl=11.5406

Epoch 18/50 | Train: total=90.9137, recon=74.1177, kl=11.1973 | Test: total=91.1020, recon=74.0196, kl=11.3883

Epoch 19/50 | Train: total=90.5212, recon=73.3722, kl=11.4327 | Test: total=90.9083, recon=74.0034, kl=11.2699

Epoch 20/50 | Train: total=90.0120, recon=72.5680, kl=11.6293 | Test: total=90.0597, recon=72.4162, kl=11.7623

Epoch 21/50 | Train: total=89.5205, recon=71.7794, kl=11.8274 | Test: total=89.5472, recon=71.8288, kl=11.8123

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

```
self._shutdown_workers()
```

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

```
if w.is_alive():Exception ignored in:
```

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

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File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

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self._shutdown_workers() ^
```

^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

```
^    ^if w.is_alive():^^
```

^ ^ ^ ^ ^ ^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

```
^ ^ ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
```

^ ^ ^ ^ ^ ^ ^ ^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

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assert self._parent_pid == os.getpid(), 'can only test a child process'^
```

```
^AssertionError^: ^can only test a child process^
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```
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^
```

```
^Traceback (most recent call last):
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^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

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self._shutdown_workers()AssertionError
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```

can only test a child process

```
if w.is_alive():
```

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

```
self._shutdown_workers() ^
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```


Epoch 23/50 | Train: total=88.9412, recon=70.9717, kl=11.9797 | Test: total=89.2611, recon=71.2019, kl=12.0394

Epoch 24/50 | Train: total=88.8147, recon=70.7580, kl=12.0378 | Test: total=89.1736, recon=70.7058, kl=12.3119

Epoch 25/50 | Train: total=88.6801, recon=70.5833, kl=12.0645 | Test: total=88.9893, recon=71.0826, kl=11.9378

Epoch 26/50 | Train: total=88.5228, recon=70.3954, kl=12.0849 | Test: total=88.9007, recon=70.8863, kl=12.0096

Epoch 27/50 | Train: total=88.4112, recon=70.1491, kl=12.1747 | Test: total=88.7786, recon=70.9659, kl=11.8751

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>Exception ignored in:
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>Traceback (most recent call last):
```

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

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File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

```
self._shutdown_workers()self._shutdown_workers()
```

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

```
if w.is_alive():if w.is_alive():
```

```
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
```

```
^    ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
```

^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

```
assert self._parent_pid == os.getpid(), 'can only test a child process'
```

```
^AssertionError: can only test a child process^
```

^Exception ignored in:


```
^AssertionError^: ^can only test a child process
```

```
AssertionError: Exception ignored in: can only test a child process<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Exception ignored in: Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

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```
self._shutdown_workers()    self._shutdown_workers()
```

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

```
if w.is_alive():
```

```
if w.is_alive():
```

```
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
```

^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

```
assert self._parent_pid == os.getpid(), 'can only test a child process'
```

```
assert self._parent_pid == os.getpid(), 'can only test a child process'
```

```
AssertionErrorAssertionError: : can only test a child processcan only test a child process
```

```
Exception ignored in: File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
```

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0><function _MultiProcessingDataLoaderIter.__del__
```

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

```
Traceback (most recent call last):
```

```
self.shutdown_workers()
```

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

```
self.shutdown_workers()
```

```
if w.is_alive():
```

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

```
if w.is_alive():
```

^^ ^ ^ ^^ ^ ^^^^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

```
^assert self._parent_pid == os.getpid(), 'can only test a child process'^^
```

^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

```
assert self._parent_pid == os.getpid(), 'can only test a child process'
```



```
assert self._parent_pid == os.getpid(), 'can only test a child process'
```

```
AssertionError: : can only test a child process
AssertionError: : can only test a child process
```

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

```
Exception ignored in: File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
```

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

```
self._shutdown_workers() File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
```

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

```
self._shutdown_workers()
```

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

```
if w.is_alive():
```

```
if w.is_alive():
```

```
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
```

^

```
assert self._parent_pid == os.getpid(), 'can only test a child process' File "/usr/lib/python3.12/multiprocessing/process.py", line 160
```

```
assert self._parent_pid == os.getpid(), 'can only test a child process'
```

^AssertionError

```
: AssertionError can only test a child process:
```

can only test a child process

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>Exception ignored in:
```

Traceback (most recent call last):

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/data
```

```
self._shutdown_workers()Traceback (most recent call last):
```

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1637, in `_shutdown_workers`

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

```
self._shutdown_workers()
```

```
if w.is_alive(): File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```

```
if w.is_alive():
```

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

^

```
assert self.parent.pid == os.getpid() 'can only test a child process' File "/usr/lib/python3.12/multiprocessing/process.py" line 160
```

```
assert self._parent_pid == os.getpid(), 'can only test a child process'  File "/usr/lib/python3.12/multiprocessing/process.py", line 100
```

```
    assert self._parent_pid == os.getpid(), 'can only test a child process'
           ^ ^ ^ ^ ^ ~~~~~
```

```
^AssertionError: ^can only test a child process^
```

```
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^
```

```
^Traceback (most recent call last):
```

```
^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
```

```
^    self._shutdown_workers()
```

```
^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```

```
AssertionError      : can only test a child processif w.is_alive():
```

```
Exception ignored in:  <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

```
Traceback (most recent call last):
```

```
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
```

```
    self._shutdown_workers()^
```

```
^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```

```
^^    ^if w.is_alive():^
```

```
^ ^ ^ ^ ^ ^
```

```
  File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
```

```
    ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
```

```
^ ^ ^ ^ ^ ^ ^ ^ ^ ^
```

```
  File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
```

```
    ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
```

```
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ~~~~~
```

```
^AssertionError: ^can only test a child process^
```

```
^^^Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^
```

```
^Traceback (most recent call last):
```

```
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
```

```
^    ^^self._shutdown_workers()^
```

```
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```

```
AssertionError      : if w.is_alive():can only test a child process
```

```
Exception ignored in:  <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>  ^^
```

```
Traceback (most recent call last):
```

```
^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
```

```
^^    self._shutdown_workers()^
```

```
^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```

```
^    ^if w.is_alive():^
```

```
^  ^
```

```
  File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
```

```
    assert self._parent_pid == os.getpid(), 'can only test a child process'
^  ^  ^  ^  ^  ^  ^  ^
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'^
^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^  ^
^^AssertionError: ^can only test a child process^^
^^^^^
```

AssertionError: can only test a child process

Epoch 28/50 | Train: total=88.2876, recon=69.9444, kl=12.2288 | Test: total=88.8994, recon=70.6234, kl=12.1840

Epoch 29/50 | Train: total=88.1530, recon=69.7070, kl=12.2973 | Test: total=88.8233, recon=70.0708, kl=12.5017

Epoch 30/50 | Train: total=88.0673, recon=69.5904, kl=12.3180 | Test: total=88.5036, recon=70.0272, kl=12.3176

Epoch 31/50 | Train: total=87.9725, recon=69.3635, kl=12.4060 | Test: total=88.5025, recon=70.3007, kl=12.1345

Epoch 32/50 | Train: total=87.8643, recon=69.2226, kl=12.4278 | Test: total=88.4278, recon=70.0027, kl=12.2834

Epoch 33/50 | Train: total=87.7375, recon=69.0158, kl=12.4811 | Test: total=88.3236, recon=69.7798, kl=12.3625

Exception ignored in: Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0><function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

Traceback (most recent call last):

self._shutdown_workers() File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__


```

AssertionError : can only test a child process^^^^
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^
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    assert self._parent_pid == os.getpid(), 'can only test a child process'
    self._shutdown_workers()
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
^ ^ ^ ^ ^ ^ ^ ^ ^
^AssertionError^: ^can only test a child process^
^Exception ignored in: ^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^
^Traceback (most recent call last):
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^^   ^^self._shutdown_workers()^^^^
^^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^^   ^^^^if w.is_alive():^
^ ^ ^ ^ ^
^AssertionError^: ^can only test a child process^
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^
Traceback (most recent call last):
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^   ^self._shutdown_workers()^
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

    if w.is_alive(): File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

    assert self._parent_pid == os.getpid(), 'can only test a child process'
        ^ ^ ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
^ ^ ^ ^ ^ ^ ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^
^AssertionError^: ^can only test a child process^
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Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
^^^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

```



```

^ if w.is_alive():^
^ ^^^^^
AssertionError : can only test a child process^^
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^
^^Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^   ^self._shutdown_workers()^
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
if w.is_alive():
assert self._parent_pid == os.getpid(), 'can only test a child process'
    ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
^ ^^ ^ ^^ ^ ^ ^ ^ ^ ^ ^ ^^^^^^^^^^^^^
^AssertionError^: ^^can only test a child process^
^Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^   ^self._shutdown_workers()
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^^   ^^if w.is_alive():^
^ ^   ^^^ ^^^^^^^^^^^^^
AssertionError^^: can only test a child process^
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
  File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
assert self._parent_pid == os.getpid(), 'can only test a child process'
  self._shutdown_workers()
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^^^^^^^^^^^^^^^^^^^^^^^^^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'
^ ^ ^^ ^ ^ ^^ ^ ^ ^^^^^
^AssertionError^: ^can only test a child process
^^^Exception ignored in: ^^^^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
   ^^self._shutdown_workers()^

```

[illegible]

```

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in __del__
^^^^    ^self._shutdown_workers()^^
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^^^    ^if w.is_alive():^
^ ^ ^ ^ ^ ^ ^ ^ ^ ^
AssertionError: can only test a child process^
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
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^^    ^self._shutdown_workers()

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    assert self._parent_pid == os.getpid(), 'can only test a child process'
if w.is_alive():
    ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^
^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^    ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^
^AssertionError^^: ^can only test a child process^^
^^^^^^^^^^^^^^^^
AssertionError: can only test a child process

```

Epoch 34/50 | Train: total=87.5551, recon=68.7518, kl=12.5355 | Test: total=87.9997, recon=69.1830, kl=12.5444

Epoch 35/50 | Train: total=87.3730, recon=68.4857, kl=12.5915 | Test: total=88.1277, recon=69.1582, kl=12.6464

Epoch 36/50 | Train: total=87.2953, recon=68.2791, kl=12.6774 | Test: total=87.8733, recon=68.7088, kl=12.7764

Epoch 37/50 | Train: total=87.1375, recon=68.0419, kl=12.7304 | Test: total=87.9665, recon=68.7653, kl=12.8008

Epoch 38/50 | Train: total=87.0732, recon=67.8485, kl=12.8165 | Test: total=87.6244, recon=68.5700, kl=12.7030

Exception ignored in: <function _M

Traceback (most recent call last):

```
Exception ignored in: File /usr/local/lib/python3.12/dist-packages
```

<Function _MultiProcessingDataLoad

Back (most recent call last):

File: /usr/local/

```
_shutdown_workers()
```

File: /usr/local/

```
shutdown workers()
```

File ~/usr

```
if w.is_active():
```

```

/e() :

```

```
parent_pid == os.getpid(), can only test a child process
```

AssertionError:

can only test a child processAssertionError

```

: Exception ignored in:

```

Traceback (most recent call last):

File ~/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py, line 1634, in __del__

```
/usr/local/lib/python3.12/dist-packages/torch/ut
```

File `/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py`, line 1637, in `_shutdown_workers`

```
/usr/local/lib/python3.12/dist-p
```



```

    if w.is_alive():
    ^ ^ ^^^^^^^^^^^^^^^^^^^^^
^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   assert self._parent_pid == os.getpid(), 'can only test a child process'^
^ ^
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
            ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^AssertionError: ^can only test a child process^
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<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>AssertionError:
can only test a child processTraceback (most recent call last):

  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>self._shutdown_workers()

  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():Traceback (most recent call last):

  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
    ^ ^ ^ ^ ^ ^ ^ ^ ^ ^^^^^^^^^^^^^
^^^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
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^
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
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^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^

Traceback (most recent call last):
AssertionError:   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
can only test a child process
self._shutdown_workers()
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
Exception ignored in:   if w.is_alive():
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers() ^

```



```
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    assert self._parent_pid == os.getpid(), 'can only test a child process'
  self._shutdown_workers()
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
      if w.is_alive():
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^
^AssertionError: can only test a child process^^
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
AssertionError: can only test a child process
```

Epoch 40/50 | Train: total=86.9050, recon=67.5808, kl=12.8828 | Test: total=87.6366, recon=68.0818, kl=13.0366

Epoch 41/50 | Train: total=86.8295, recon=67.4549, kl=12.9164 | Test: total=87.5595, recon=67.8878, kl=13.1145

Epoch 42/50 | Train: total=86.7648, recon=67.3327, kl=12.9548 | Test: total=87.7580, recon=68.4470, kl=12.8740

Epoch 43/50 | Train: total=86.6758, recon=67.2262, kl=12.9664 | Test: total=87.5131, recon=68.1309, kl=12.9215

Epoch 44/50 | Train: total=86.5804, recon=67.0373, kl=13.0288 | Test: total=87.3231, recon=67.7491, kl=13.0494

Epoch 45/50 | Train: total=86.4772, recon=66.8220, kl=13.1035 | Test: total=87.0820, recon=67.3402, kl=13.1612

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):

```

Traceback (most recent call last):
Exception ignored in: File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
self._shutdown_workers()
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
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    ^^ ^   ^ ^ ~~~~~~
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^    ^assert self._parent_pid == os.getpid(), 'can only test a child process'^
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    ^ ^ ^ ^ ^ ^ ^ ~~~~~~
AssertionError^: ^can only test a child process^
```



```
Epoch 49/50 | Train: total=86.0297, recon=65.9619, kl=13.3785 | Test: total=86.8756, recon=67.1290, kl=13.1644
```

Epoch 50/50 | Train: total=85.8540, recon=65.7183, kl=13.4238 | Test: total=86.7765, recon=66.7700, kl=13.3377

Training Convolutional VAE...

Epoch 01/50 | Train: total=128.0009, recon=113.2258, kl=9.8501 | Test: total=103.4562, recon=86.1203, kl=11.5573

```
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```

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^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
AssertionError: can only test a child process
Epoch 02/50 | Train: total=99.6033, recon=81.8296, kl=11.8491 | Test: total=95.1989, recon=74.8917, kl=13.5381

Epoch 03/50 | Train: total=93.5308, recon=73.4248, kl=13.4040 | Test: total=91.6034, recon=69.8669, kl=14.4910

Epoch 04/50 | Train: total=90.7273, recon=69.7933, kl=13.9560 | Test: total=89.3036, recon=68.0967, kl=14.1379

Epoch 05/50 | Train: total=89.0461, recon=67.6685, kl=14.2517 | Test: total=88.4056, recon=66.7176, kl=14.4587

Epoch 06/50 | Train: total=88.5374, recon=66.8803, kl=14.4381 | Test: total=87.9416, recon=66.1966, kl=14.4967

Epoch 07/50 | Train: total=87.9836, recon=66.0780, kl=14.6037 | Test: total=87.3988, recon=65.3968, kl=14.6680

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[illegible]


```

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```


Epoch 09/50 | Train: total=87.3677, recon=65.2490, kl=14.7458 | Test: total=86.8027, recon=64.8996, kl=14.6021

Epoch 10/50 | Train: total=87.1597, recon=64.9100, kl=14.8332 | Test: total=86.6235, recon=64.6397, kl=14.6559

Epoch 11/50 | Train: total=87.0071, recon=64.7193, kl=14.8585 | Test: total=86.6149, recon=63.8948, kl=15.1468

Epoch 12/50 | Train: total=86.6839, recon=64.2270, kl=14.9713 | Test: total=86.2207, recon=63.7908, kl=14.9533

Epoch 13/50 | Train: total=86.4477, recon=63.9214, kl=15.0175 | Test: total=86.1303, recon=63.9604, kl=14.7800

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[illegible]

Epoch 15/50 | Train: total=86.2347, recon=63.4266, kl=15.2054 | Test: total=85.7831, recon=62.7907, kl=15.3283

Epoch 16/50 | Train: total=86.0815, recon=63.1650, kl=15.2777 | Test: total=86.2107, recon=63.5583, kl=15.1016

Epoch 17/50 | Train: total=85.9820, recon=62.9753, kl=15.3378 | Test: total=85.8842, recon=62.3868, kl=15.6649

Epoch 18/50 | Train: total=85.7166, recon=62.6467, kl=15.3799 | Test: total=85.8999, recon=62.4747, kl=15.6168

Epoch 19/50 | Train: total=85.7082, recon=62.5615, kl=15.4311 | Test: total=85.4594, recon=62.5425, kl=15.2779

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^   ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^AssertionError^: ^can only test a child process^^
^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^     self._shutdown_workers()^
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():^
        ^ ^^^^^^^^
^AssertionError: ^^can only test a child process^
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>

Traceback (most recent call last):
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    assert self._parent_pid == os.getpid(), 'can only test a child process'self._shutdown_workers()

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
```

```

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^ ^ ~~~~~
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^     ^assert self._parent_pid == os.getpid(), 'can only test a child process'
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ~~~~~
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^^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    ^self._shutdown_workers()
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
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AssertionError:
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    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^     ^if w.is_alive():

    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
        assert self._parent_pid == os.getpid(), 'can only test a child process'
        ^^ ^^ ^ ^^ ^ ^ ^ ^ ^^
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^     ^^assert self._parent_pid == os.getpid(), 'can only test a child process'^
^   ^^ ^   ^ ^ ^ ^ ^   ~~~~~
^AssertionError^: ^can only test a child process
^^^^^^^^^^^^
AssertionError: can only test a child process
Epoch 20/50 | Train: total=85.5955, recon=62.3720, kl=15.4823 | Test: total=85.4573, recon=62.0610, kl=15.5975

Epoch 21/50 | Train: total=85.4069, recon=62.1526, kl=15.5029 | Test: total=85.3350, recon=62.0848, kl=15.5002

Epoch 22/50 | Train: total=85.3303, recon=62.0603, kl=15.5134 | Test: total=84.8797, recon=61.7347, kl=15.4301

```


Epoch 23/50 | Train: total=85.1730, recon=61.8640, kl=15.5393 | Test: total=84.9721, recon=61.6410, kl=15.5541

Epoch 24/50 | Train: total=85.1423, recon=61.7930, kl=15.5662 | Test: total=85.3432, recon=62.3672, kl=15.3173

Epoch 25/50 | Train: total=85.0570, recon=61.6118, kl=15.6302 | Test: total=85.6518, recon=62.3796, kl=15.5148

```
Exception ignored in: Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    Exception ignored in:   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
self._shutdown_workers()    <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>

self._shutdown_workers()Traceback (most recent call last):

  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    self._shutdown_workers()
if w.is_alive():
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    Exception ignored in:  if w.is_alive():if w.is_alive():

<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
  ^  ^ self._shutdown_workers()^^
^^  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^ ^^^    ^^^^^if w.is_alive():^^^^^^
^^ ^^^ ^
^ ^  File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^ ^    ^^assert self._parent_pid == os.getpid(), 'can only test a child process' ^
```


[illegible]

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^assert self._parent_pid == os.getpid(), 'can only test a child process'^
^
if w.is_alive(): ^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^
assert self._parent_pid == os.getpid(), 'can only test a child process' File "/usr/lib/python3.12/multiprocessing/process.py", line
    assert self._parent_pid == os.getpid(), 'can only test a child process'
        ^ ^      ^^      ^^^ ^^^ ^^^ ^^^ ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
^^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^^     ^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^
^^^^ ^^^^^^^ ^^^ ^^ ^^^^^ ^^^^^ ^^ ^^^ ^^^ ^^^
^^ ^AssertionError^^: ^^^^^can only test a child process^^
^^^^^^^^^^Exception ignored in: ^^^^^
^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^AssertionError
^: Traceback (most recent call last):

^can only test a child process File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
AssertionError^
   : self._shutdown_workers()^can only test a child processException ignored in:
^
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/data
^
Exception ignored in: if w.is_alive():^Traceback (most recent call last):
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

    ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> ^ self._shutdown_workers()
^Traceback (most recent call last):

^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^   ^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^if w.is_alive(): ^
^self._shutdown_workers()^ ^^ ^
^ ^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^ ^     ^^if w.is_alive(): ^
^^ ^
^^AssertionError ^^: ^^can only test a child process ^^
^^
^ Exception ignored in: File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^ <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^
^assert self._parent_pid == os.getpid(), 'can only test a child process'^Traceback (most recent call last):
```



```

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^assert self._parent_pid == os.getpid(), 'can only test a child process' ^^
^ ^ self._shutdown_workers()^
^ ^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
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if w.is_alive():^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
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^ ^ ^ assert self._parent_pid == os.getpid(), 'can only test a child process' ^^
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^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^ ^^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^
^ ^^^ ^^^ ^^^ ^^^ ^^^ ^^^ ^^^ ^^^ ^^^
^^ ^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^ ^^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^^^
^^^^ ^^^^^ ^^^^^ ^^^^^ ^^^^^ ^^^^^ ^^^
^AssertionError^^^^: can only test a child process^
^^ ^^ Exception ignored in: ^^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^
^Traceback (most recent call last):
^ ^^^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^
^^ ^AssertionErrorself._shutdown_workers(): ^^can only test a child process

^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^^^Exception ignored in: ^^ ^^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^if w.is_alive():
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File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
self._shutdown_workers() ^ ^^^
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File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
AssertionError ^: ^ can only test a child process^if w.is_alive():
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^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
self._shutdown_workers()^
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^^^ ^if w.is_alive():^^^^
^^^^ ^^
^ ^AssertionError^ : ^^ can only test a child process^ ^
^
^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
Exception ignored in: ^^ <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^assert self._parent_pid == os.getpid()

```



```

^ ^Traceback (most recent call last):
^ ^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^^ ^    ^self._shutdown_workers()

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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   assert self._parent_pid == os.getpid(), 'can only test a child process'
^if w.is_alive(): assert self._parent_pid == os.getpid(), 'can only test a child process'
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^ ^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
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^^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^
^ ^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^ ^   ^ assert self._parent_pid == os.getpid(), 'can only test a child process'^^^^
^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^ ^^
^^^^AssertionError^^
^: AssertionError^^can only test a child process: ^^
can only test a child process^^
^^Exception ignored in: ^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^Exception ignored in: ^
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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
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^^

^if w.is_alive(): File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
AssertionError
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^
^ ^ Exception ignored in: ^ <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^ ^
^ ^Traceback (most recent call last):
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^^^^AssertionError self._shutdown_workers(): ^
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^ ^   ^can only test a child process^^if w.is_alive():
^^

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^^^Exception ignored in:
^^ <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> File "/usr/lib/python3.12/multiprocessing/process.py", line 160,

Traceback (most recent call last):
assert self._parent_pid == os.getpid(), 'can only test a child process'
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^   ^   self._shutdown_workers()

    ^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    ^   if w.is_alive():
assert self._parent_pid == os.getpid(), 'can only test a child process'
^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^ ^
    ^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^   ^ ^assert self._parent_pid == os.getpid(), 'can only test a child process'^^
^^^ ^^ ^^^ ^^^^ ^^^^ ^^^^ ^^^^
    ^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^^   ^ ^assert self._parent_pid == os.getpid(), 'can only test a child process'^^^
^^   ^   ^ ^^^^^^ ^^^ ^ ^^^ ^^^ ^^^^ ^^^^^^^^^^ ^^^^^^^^^^^^^

^^AssertionErrorAssertionError: : ^^can only test a child process^can only test a child process^
^
^^Exception ignored in: ^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>Exception ignored in:
^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^Traceback (most recent call last):

^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^^   ^Traceback (most recent call last):
^^self._shutdown_workers() File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^   ^self._shutdown_workers()^^^^^^
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^if w.is_alive():
    ^^
    ^if w.is_alive():AssertionError ^
        : ^
^ can only test a child process^^ ^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^ ^^
^^Traceback (most recent call last):
^^^^^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^^   ^^^self._shutdown_workers()^^
^
    File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
AssertionError^^: ^   ^can only test a child process^

```

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^if w.is_alive():^Exception ignored in:

^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> File "/usr/lib/python3.12/multiprocessing/process.py", line 160,

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
assert self._parent_pid == os.getpid(), 'can only test a child process'assert self._parent_pid == os.getpid(), 'can only test a child pr

^ self._shutdown_workers() ^
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^ ^ ^if w.is_alive():^
^ ^ ^ ^ ^ ^
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^ ^ ^^^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^^^
^^^^ ^ ^^^^ ^^^ ^^^ ^^^ ^^^ ^^^ ^^^ ^^^
^^ File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^ ^ ^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^^
^^ ^^^^ ^ ^^^^^ ^ ^^^ ^^^ ^^^^^ ^^^ ^^^^^ ^^^
^^^AssertionError^^: ^^can only test a child process^^
^
^^AssertionError^^^^^^Exception ignored in: : ^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>can only test a child

Traceback (most recent call last):
^^^^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^ ^
^^Traceback (most recent call last):
^self._shutdown_workers()
^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^^^ ^if w.is_alive(): ^
self._shutdown_workers()^^ ^^

^AssertionError ^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^: can only test a child process^
^^^^^if w.is_alive():^Exception ignored in:
^^
^AssertionError<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> ^:
^ Traceback (most recent call last):
can only test a child process ^ ^
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^Exception ignored in: ^self._shutdown_workers()<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>~

```


Epoch 29/50 | Train: total=84.7991, recon=61.2583, kl=15.6938 | Test: total=84.6307, recon=61.0402, kl=15.7270

Epoch 30/50 | Train: total=84.8054, recon=61.2460, kl=15.7063 | Test: total=85.1453, recon=60.8574, kl=16.1919

Epoch 31/50 | Train: total=84.7605, recon=61.1841, kl=15.7176 | Test: total=84.3760, recon=61.1342, kl=15.4946

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

self._shutdown_workers()

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

if w.is_alive():

^^^^^^^^^^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(), 'can only test a child process'

^^

AssertionError: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

self._shutdown_workers()

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

if w.is_alive():

^^^^^^^^^^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(), 'can only test a child process'

^^

AssertionError: can only test a child process

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>

Traceback (most recent call last):

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__

self._shutdown_workers()

File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

if w.is_alive():

^^^^^^^^^^

File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive

assert self._parent_pid == os.getpid(). 'can only test a child process'


```
File ~/usr/lib/python3.12/multiprocessing/process.py , line 100, in _active
    assert self._parent_pid == os.getpid(), 'can only test a child process'
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
AssertionError: can only test a child process
```

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

AssertionError: can only test a child process

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

AssertionError: can only test a child process

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^^^^^^^^^^^^^
```

```
File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
AssertionError: can only test a child process
```

Exception ignored in:

```
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>Traceback (most recent call last):
```

```
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
Exception ignored in: <_C10DistributedDataStore object at 0x7f2e00000000>
```

```
File: //usr/local/lib/python3.12/dist-packages/tables/tables1codes.xml:1637:is:abstract:yes:yes
```



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^^      ^^self._shutdown_workers()self._shutdown_workers()^^
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File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^   ^^   if w.is_alive():if w.is_alive():^^

^^ ^^ ^
^ AssertionError ^: ^ can only test a child process ^^ ^
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assert self._parent_pid == os.getpid(), 'can only test a child process' File "/usr/local/lib/python3.12/dist-packages/torch/utils/da

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^^^^^^    ^^assert self._parent_pid == os.getpid(), 'can only test a child process'^^^^
^^ ^^^^
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^ ^ ^^    ^^    ^ ^ ^    ^^ ^^^    ^^ ^^ ^    ^^ ^^^^^^^^^^^^^^^^^
^^^^AssertionErrorAssertionError: ^: can only test a child process^^can only test a child process
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^^^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^^^
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^   ^^^^^
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exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> self._shutdown_workers()
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  AssertionError: if w.is_alive():
    can only test a child process
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^   ^   ^   ^   ^
^   ^   ^   ^   ^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^   ^   ^^ assert self._parent_pid == os.getpid(), 'can only test a child process'^
^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^
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^AssertionError^^^^^^: ^^^^^can only test a child process
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^   ^ assert self._parent_pid == os.getpid(), 'can only test a child process' ^ ^
^ ^ ^^ ^ ^ ^^ ^ ^^   ^^ ^^   ^^ ^^ ^^ ^ ~~~~~
^AssertionError
^^AssertionError: ^^: can only test a child process^^
can only test a child process^^

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can only test a child process
^^Exception ignored in: ^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^
^^Exception ignored in: ^^<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^^
^^^Traceback (most recent call last):
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^^ ^^ ^^ ^^ ^ ^^ ^^ ^ ^^
AssertionError^^: can only test a child process^
^^^Exception ignored in:

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Exception ignored in:
^AssertionError<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>^:
^Traceback (most recent call last):
can only test a child process^ File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
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^^Traceback (most recent call last):
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^^AssertionError
  : ^can only test a child processself._shutdown_workers()AssertionError: ^

can only test a child process^
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^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
^       self._shutdown_workers()^self._shutdown_workers() ^

^^   File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
^^    ^if w.is_alive():^^    ^
^if w.is_alive():^ ^ ^
^ ^ ^ ^ ^ ^
  ^AssertionError ^ : ^^ can only test a child process^ ^

^^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^^^    ^Exception ignored in: ^^assert self._parent_pid == os.getpid(), 'can only test a child process'<function _MultiProcessingDataL
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  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers

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    assert self._parent_pid == os.getpid(), 'can only test a child process'    if w.is_alive():

    assert self._parent_pid == os.getpid(), 'can only test a child process'
        ^    ^    ^    ^    ^^    ^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^    ^^
^^   File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
^^^^    ^^assert self. parent pid == os.getpid(). 'can only test a child process'^^

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^^^^ ^^^^^ ^^^^^ ^^^ ^
^^ AssertionError^^ ^: ^^ can only test a child process
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AssertionError: can only test a child process
Epoch 32/50 | Train: total=84.6169, recon=61.0534, kl=15.7090 | Test: total=84.5005, recon=61.3664, kl=15.4227

Epoch 33/50 | Train: total=84.5281, recon=60.9208, kl=15.7382 | Test: total=84.6978, recon=60.7656, kl=15.9548

Epoch 34/50 | Train: total=84.6324, recon=61.0146, kl=15.7452 | Test: total=84.9178, recon=60.9489, kl=15.9793

Epoch 35/50 | Train: total=84.4534, recon=60.8220, kl=15.7542 | Test: total=84.5047, recon=61.0256, kl=15.6527

Epoch 36/50 | Train: total=84.5075, recon=60.8414, kl=15.7774 | Test: total=84.2277, recon=60.5357, kl=15.7947

Epoch 37/50 | Train: total=84.4916, recon=60.7811, kl=15.8071 | Test: total=84.1745, recon=60.5582, kl=15.7442

Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
Traceback (most recent call last):
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
    self._shutdown_workers()
  File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    if w.is_alive():
        ^^^^^^^^^^^
  File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
```



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File "/usr/lib/python3.12/multiprocessing/process.py", line 160, in is_alive
    assert self._parent_pid == os.getpid(), 'can only test a child process'
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[illegible]

[illegible]

[illegible]

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        ^      ^      ^      ^      ^      ^      ^      ^      ^      ^      ^      ^      ^      ^      ^      ^
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can only test a child process^AssertionError
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AssertionError: self._shutdown_workers(): can only test a child process

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^Traceback (most recent call last):
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^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^
^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^   ^
^^AssertionError^^: ^^can only test a child process^^
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AssertionError: ^can only test a child process^
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AssertionError: can only test a child process
Epoch 38/50 | Train: total=84.3999, recon=60.6801, kl=15.8132 | Test: total=86.7269, recon=62.5515, kl=16.1169

Epoch 39/50 | Train: total=84.4313, recon=60.6751, kl=15.8375 | Test: total=84.1349, recon=60.6188, kl=15.6774

Epoch 40/50 | Train: total=84.3263, recon=60.5418, kl=15.8563 | Test: total=84.1348, recon=60.3365, kl=15.8655

```

Epoch 41/50 | Train: total=84.2441, recon=60.4873, kl=15.8379 | Test: total=84.4473, recon=60.5993, kl=15.8987

Epoch 42/50 | Train: total=84.1881, recon=60.4372, kl=15.8340 | Test: total=84.0703, recon=60.6374, kl=15.6219

Epoch 43/50 | Train: total=84.2423, recon=60.4055, kl=15.8912 | Test: total=84.3648, recon=60.6230, kl=15.8278

```
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0>
```

Traceback (most recent call last):

File `"/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py"`, line 1654, in `__del__`

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: Exception ignored in: self._shutdown_workers()can only test a child processException ignored in: <function _MultiProcessingDataLoaderI
<function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/data

Traceback (most recent call last):
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
Traceback (most recent call last):
Exception ignored in: <function _MultiProcessingDataLoaderIter.__del__ at 0x7e3eb55d7ec0> if w.is_alive(): File "/usr/local/lib/pyth
self._shutdown_workers()

Traceback (most recent call last):
File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1654, in __del__
self._shutdown_workers() File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdown_workers
    self._shutdown_workers() File "/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py", line 1637, in _shutdov

```

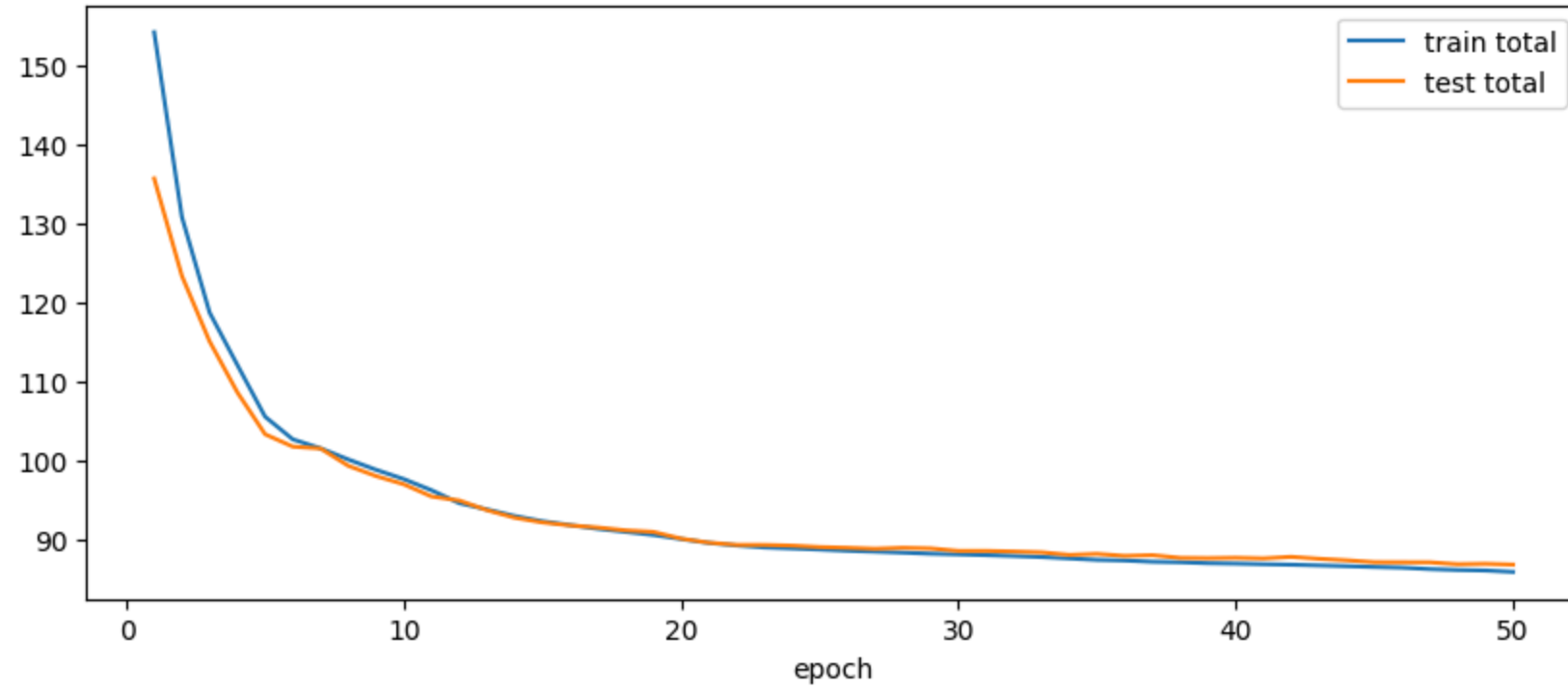
[illegible]

[illegible]

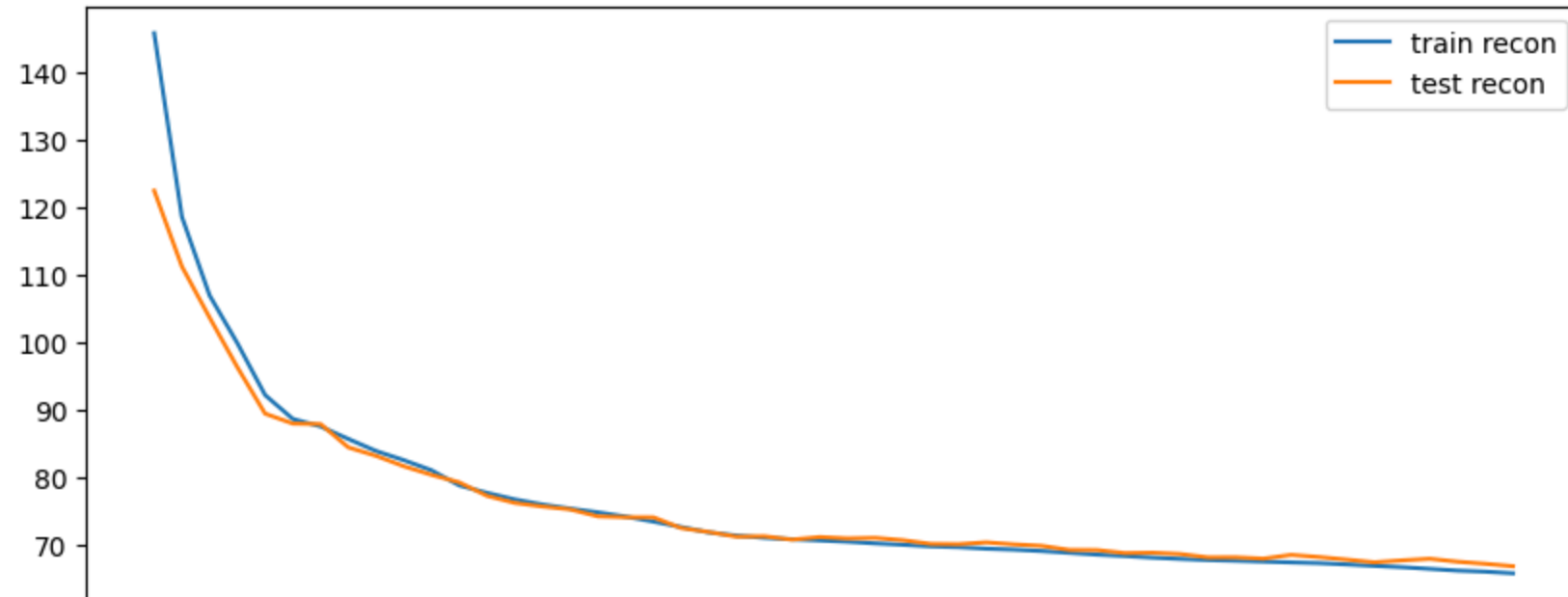
[illegible]

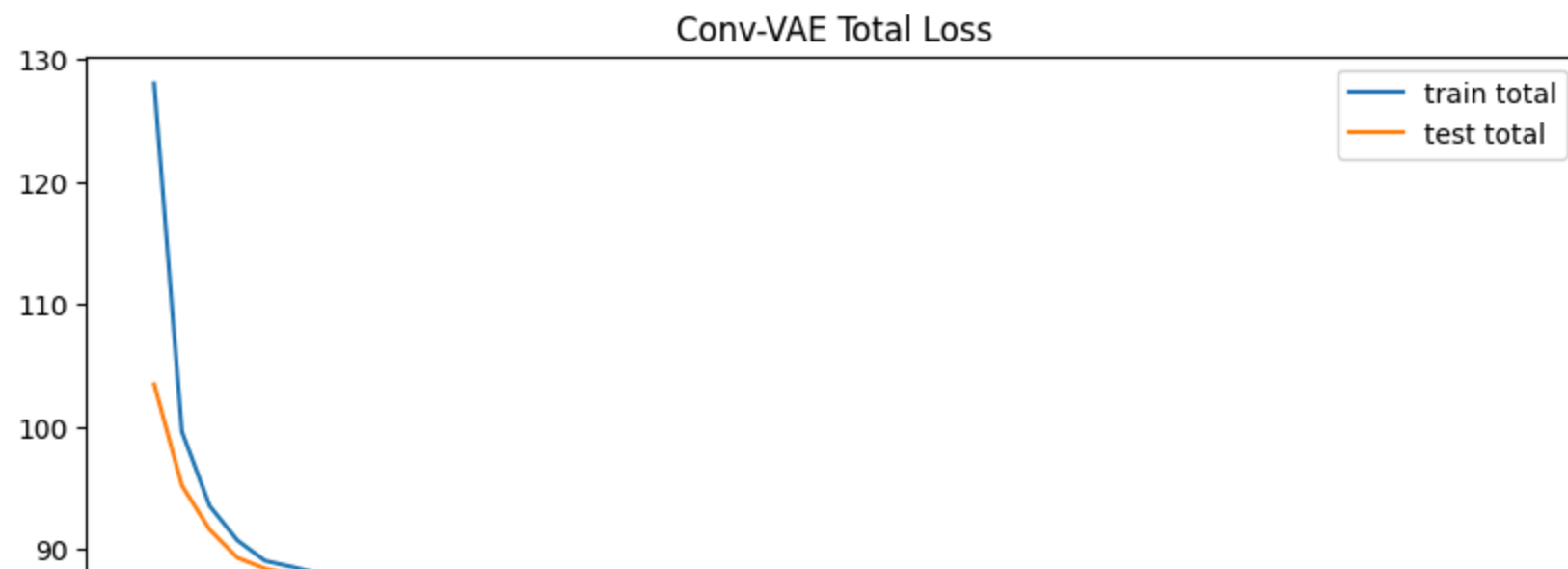
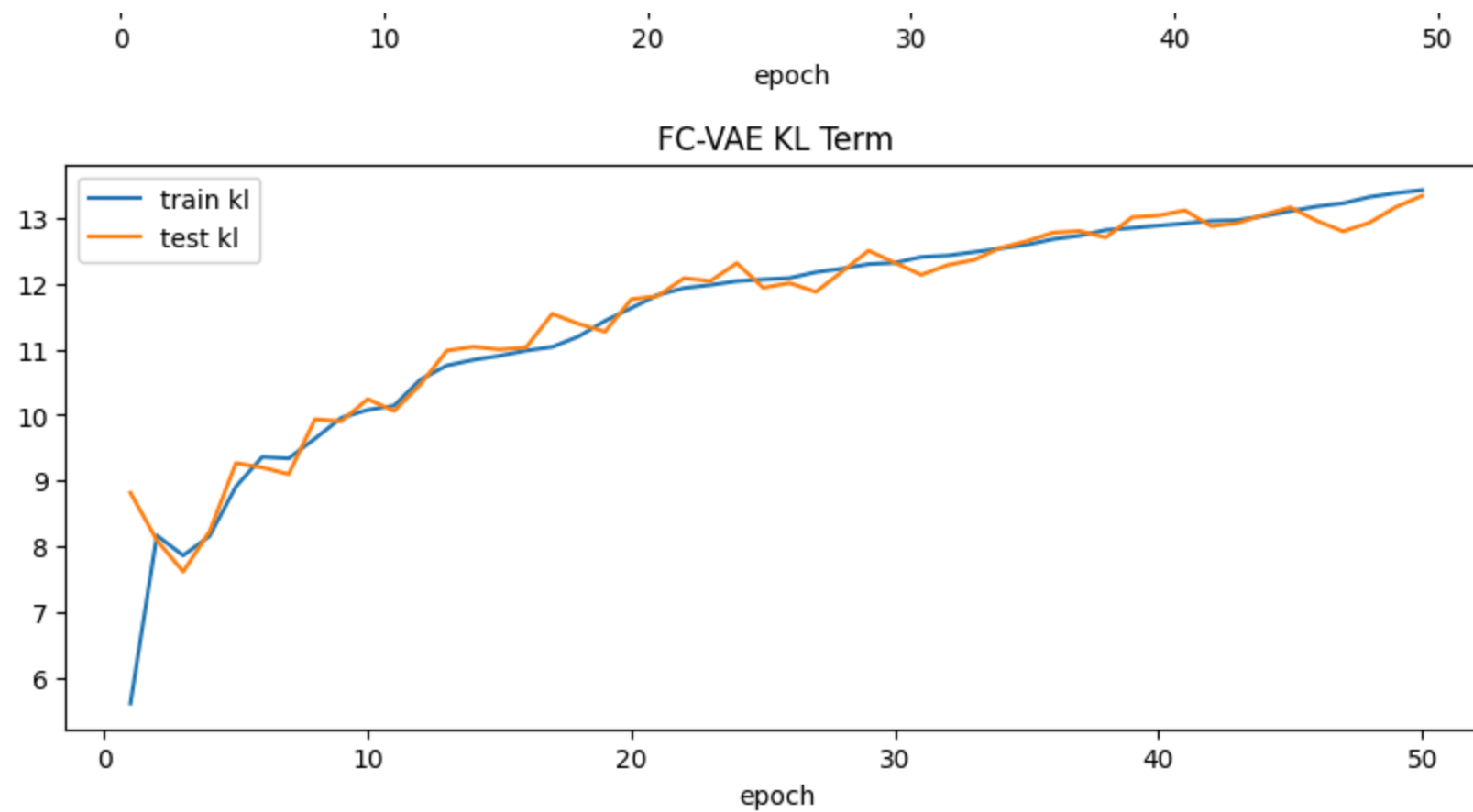

```
1 ### Cell 1.8 – Plot Loss Curves (Optional but Nice)
2 def plot_history(hist, title_prefix=""):
3     xs = np.arange(1, len(hist["train_total"]) + 1)
4
5     plt.figure(figsize=(10,4))
6     plt.plot(xs, hist["train_total"], label="train total")
7     plt.plot(xs, hist["test_total"], label="test total")
8     plt.title(f"{title_prefix} Total Loss")
9     plt.xlabel("epoch")
10    plt.legend()
11    plt.show()
12
13    plt.figure(figsize=(10,4))
14    plt.plot(xs, hist["train_recon"], label="train recon")
15    plt.plot(xs, hist["test_recon"], label="test recon")
16    plt.title(f"{title_prefix} Reconstruction Loss")
17    plt.xlabel("epoch")
18    plt.legend()
19    plt.show()
20
21    plt.figure(figsize=(10,4))
22    plt.plot(xs, hist["train_kl"], label="train kl")
23    plt.plot(xs, hist["test_kl"], label="test kl")
24    plt.title(f"{title_prefix} KL Term")
25    plt.xlabel("epoch")
26    plt.legend()
27    plt.show()
28
29 plot_history(hist_fc, "FC-VAE")
30 plot_history(hist_conv, "Conv-VAE")
31
```

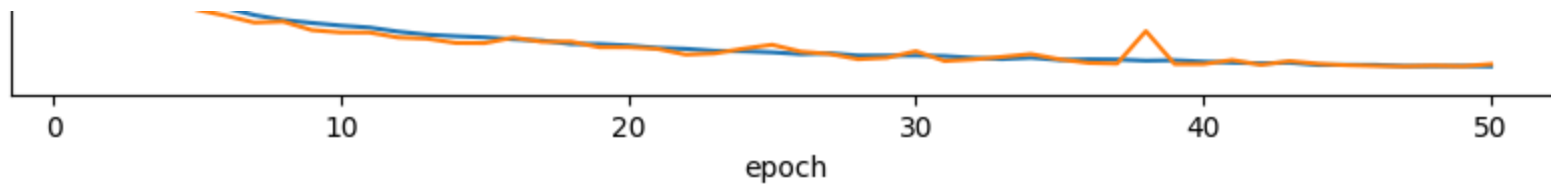

FC-VAE Total Loss



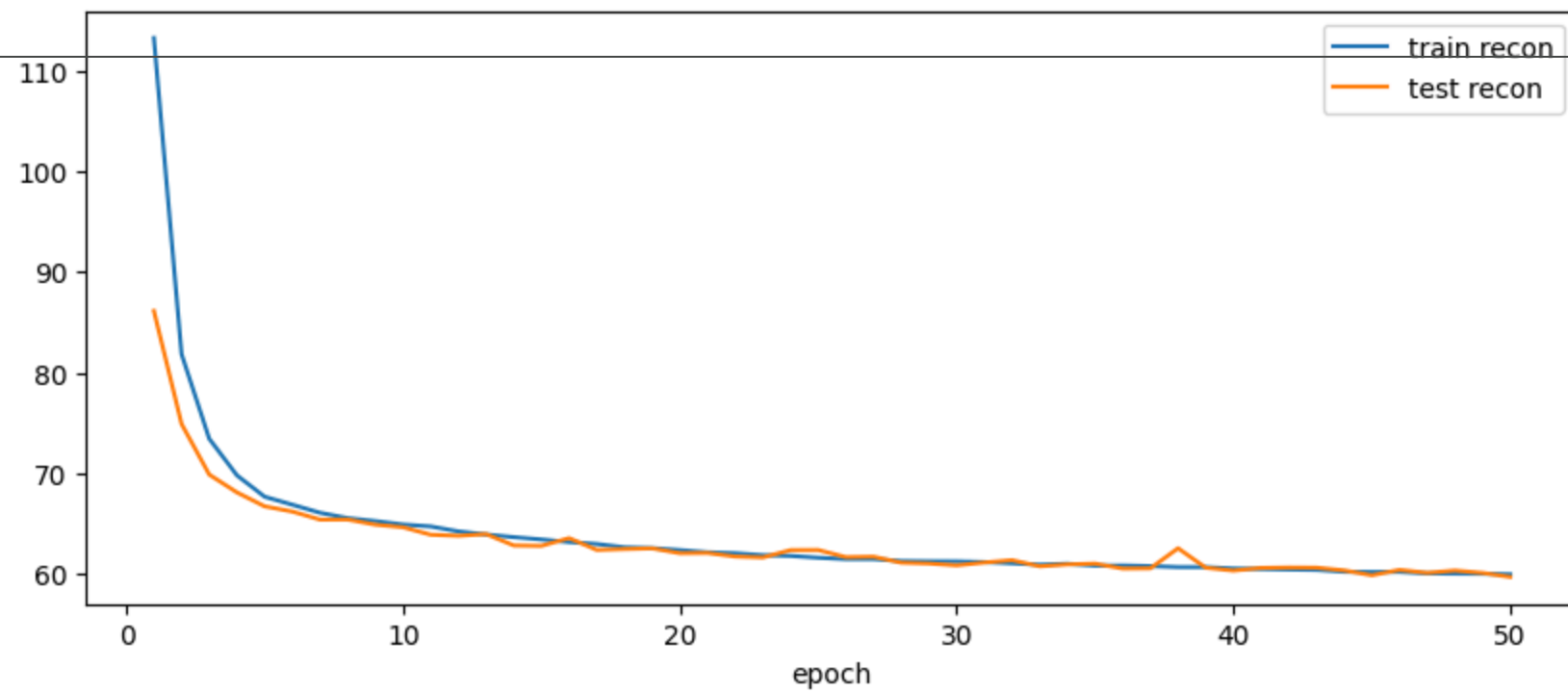
FC-VAE Reconstruction Loss



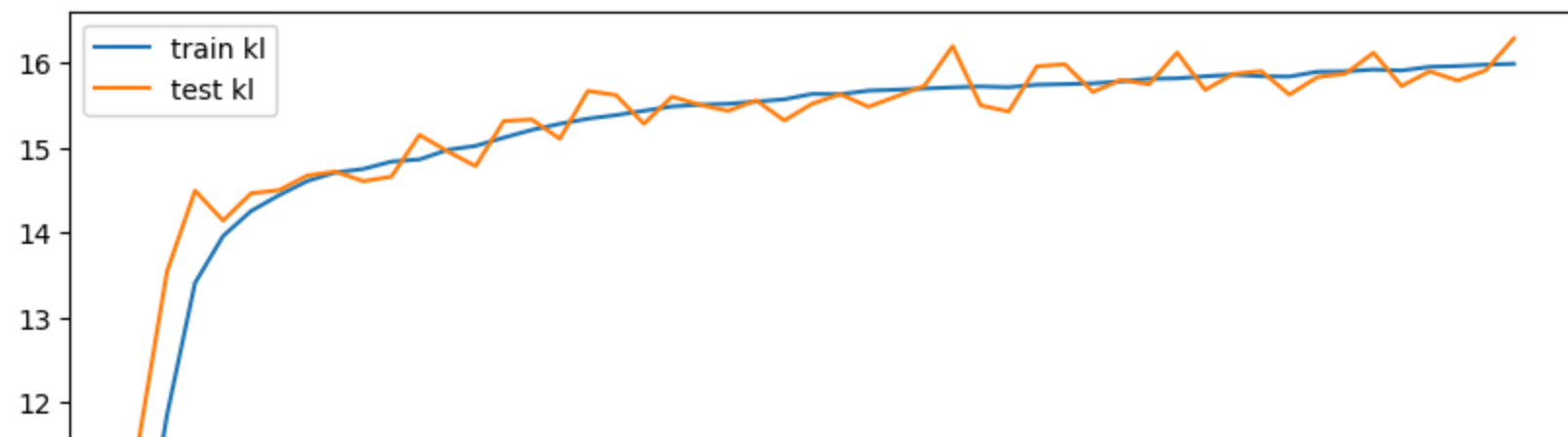




Conv-VAE Reconstruction Loss

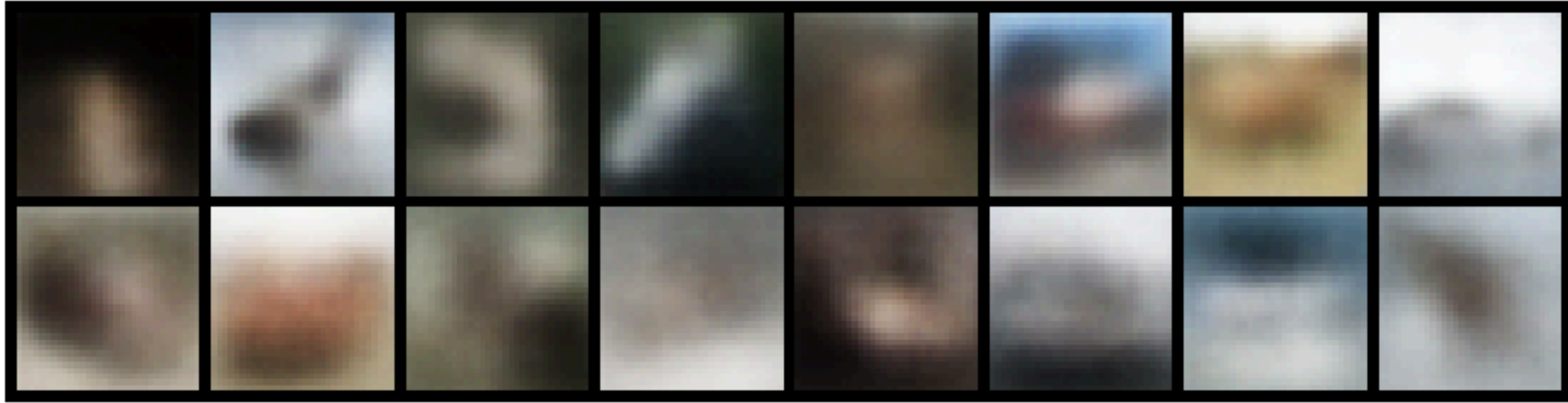


Conv-VAE KL Term

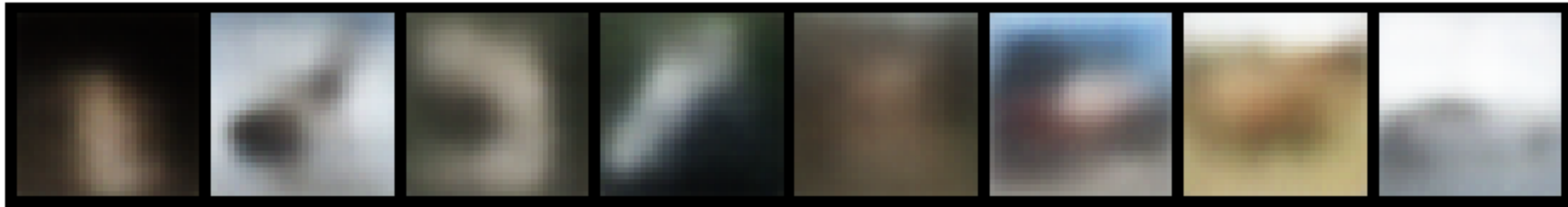


```
1 ##### Cell 1.9 – Compare Generated
2 n = 8 # show 8 samples per model row
3
4 samples_conv = sample_images(conv_vae.to(device), LATENT_DIM, n=n)
5 samples_fc    = sample_images(fc_vae.to(device),  LATENT_DIM, n=n)
6
7 # 2 rows: top=ConvVAE, bottom=FCVAE
8 combined = torch.cat([samples_conv, samples_fc], dim=0)
9 show_grid(combined, title="Generated samples: Conv-VAE (top) vs FC-VAE (bottom)", nrow=n, figsize=(12,4))
10
11 show_grid(samples_conv, title="Conv-VAE: generated samples", nrow=n, figsize=(12,2.5))
12 show_grid(samples_fc,  title="FC-VAE: generated samples",  nrow=n, figsize=(12,2.5))
13
```

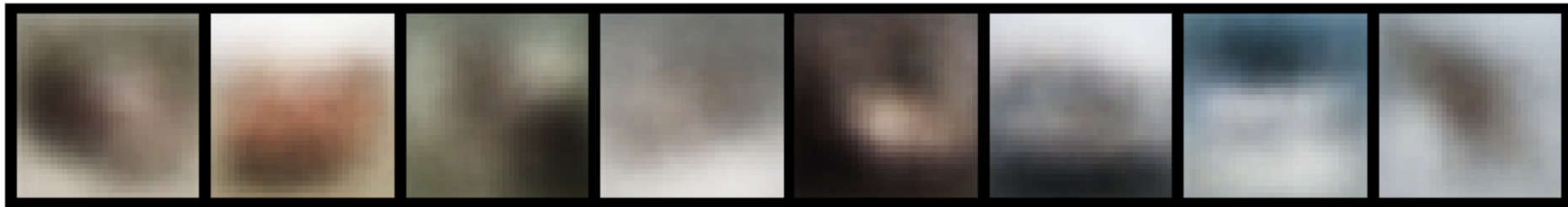
Generated samples: Conv-VAE (top) vs FC-VAE (bottom)



Conv-VAE: generated samples



FC-VAE: generated samples



```
1 ##### Cell 1.10 – Compare Reconstructions
2 x, _ = next(iter(test_loader))
3 x = x[:8].to(device)
4
5 recon_conv = reconstruct_images(conv_vae.to(device), x)
```

```

6 recon_fc = reconstruct_images(fc_vae.to(device), x)
7
8 # 3 rows: original, conv recon, fc recon
9 triplet = torch.cat([x, recon_conv, recon_fc], dim=0)
10 show_grid(triplet, title="Reconstructions: original (top), Conv-VAE (middle), FC-VAE (bottom)", nrow=8, figsize=(12,5))
11

```



Results interpretation

After implementing a convolutional encoder/decoder to replace the fully connected Variational Autoencoder (VAE) and training both models on the CIFAR-10 dataset, we found promising results that reinforce the idea that convolutional layers can effectively capture spatial relationships, thereby enhancing image quality. The Convolutional VAE (Conv-VAE) demonstrated a significantly lower reconstruction loss—and overall loss—compared to the Fully Connected VAE (FC-VAE),

while both models maintained closely aligned training and testing curves. This indicates that the learning process was stable and showed no clear signs of overfitting.

Visually, the improvements were apparent in the reconstruction comparisons: the Conv-VAE was better at preserving the coarse structure of objects, edges, and spatial layouts, whereas the FC-VAE reconstructions tended to be more washed out and less coherent. This is understandable, as flattening images early in the network can lead to a loss of locality.

Although both models produced somewhat blurry samples after 10 epochs—something commonly observed with VAEs using pixel-wise reconstruction objectives—the Conv-VAE samples exhibited a more structured and varied appearance. This suggests that convolutional feature hierarchies offer a stronger inductive bias for modeling images, which is likely to lead to higher-quality generations compared to a purely fully connected architecture. Overall, these findings highlight the advantages of convolutional layers in image processing tasks and suggest avenues for further exploration and improvement in VAE design.

✓ Task2

Using the trained VAE, interpolate between two images in the latent space and generate intermediate images. This demonstrates how smoothly the model can transition between different data points. Visualize and display the results, showing the interpolated images in a grid format to observe the transformation.

References / Citations

rishabkr. (n.d.). AutoEncoders [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/rishabkr/AutoEncoders>

evgiz. (n.d.). variational-autoencoder [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/evgiz/variational-autoencoder>

Double-click (or enter) to edit

```
1 ### Cell 2.1 Helpers (pick endpoints, lerp/slerp, interpolate + visualize)
2 import torch
```

```

3 import numpy as np
4 from torchvision.utils import make_grid
5 import matplotlib.pyplot as plt
6
7 # CIFAR-10 label names (optional, for nicer titles)
8 CIFAR10_CLASSES = [
9     "airplane", "automobile", "bird", "cat", "deer",
10    "dog", "frog", "horse", "ship", "truck"
11 ]
12
13 def show_grid(images, title="", nrow=8, figsize=(12, 3)):
14     """images: Tensor (B,3,32,32) in [0,1]"""
15     grid = make_grid(images.cpu(), nrow=nrow, padding=2)
16     plt.figure(figsize=figsize)
17     plt.imshow(grid.permute(1,2,0))
18     plt.axis("off")
19     plt.title(title)
20     plt.show()
21
22 def pick_two_images_different_classes(dataset, seed=42):
23     """Pick two random images from dataset with different class labels."""
24     rng = np.random.default_rng(seed)
25     idx1 = int(rng.integers(0, len(dataset)))
26     x1, y1 = dataset[idx1]
27     idx2 = idx1
28     while idx2 == idx1:
29         idx2 = int(rng.integers(0, len(dataset)))
30     x2, y2 = dataset[idx2]
31     if y2 != y1:
32         break
33     return (x1, y1, idx1), (x2, y2, idx2)
34
35 def lerp(z1, z2, t):
36     """Linear interpolation: (1-t)z1 + t z2"""
37     return (1 - t) * z1 + t * z2
38
39 def slerp(z1, z2, t, eps=1e-7):
40     """
41     Spherical linear interpolation.

```

```

42     Often gives smoother transitions for latent vectors on/near a hypersphere.
43     """
44     # Normalize
45     z1n = z1 / (z1.norm(dim=1, keepdim=True) + eps)
46     z2n = z2 / (z2.norm(dim=1, keepdim=True) + eps)
47
48     dot = (z1n * z2n).sum(dim=1, keepdim=True).clamp(-1.0, 1.0)
49     omega = torch.acos(dot)          # angle
50     sin_omega = torch.sin(omega).clamp_min(eps)
51
52     # If vectors are too close, fallback to lerp
53     close = (omega < 1e-3).float()
54     part1 = torch.sin((1 - t) * omega) / sin_omega
55     part2 = torch.sin(t * omega) / sin_omega
56     z = part1 * z1 + part2 * z2
57     z_l = lerp(z1, z2, t)
58
59     return close * z_l + (1 - close) * z
60
61 @torch.no_grad()
62 def interpolate_latent(model, x1, x2, steps=10, method="lerp", use_mu=True, device="cuda"):
63     """
64     model: trained VAE with .encode() and .decode()
65     x1, x2: Tensors (3,32,32) in [0,1]
66     steps: number of frames INCLUDING endpoints
67     method: "lerp" or "slerp"
68     use_mu: if True, use encoder mean as latent (deterministic, smoother);
69             if False, sample using reparameterization (more stochastic)
70     """
71     model.eval()
72     x = torch.stack([x1, x2]).to(device)
73
74     mu, logvar = model.encode(x)
75
76     if use_mu:
77         z1, z2 = mu[0:1], mu[1:2]
78     else:
79         if hasattr(model, "reparameterize"):
80             z = model.reparameterize(mu, logvar)

```

```

81         else:
82             std = torch.exp(0.5 * logvar)
83             z = mu + torch.randn_like(std) * std
84             z1, z2 = z[0:1], z[1:2]
85
86         ts = torch.linspace(0, 1, steps, device=device).view(-1, 1) # (steps,1)
87
88         if method.lower() == "slerp":
89             zs = torch.cat([slerp(z1, z2, t=ts[i:i+1]) for i in range(steps)], dim=0)
90         else:
91             zs = lerp(z1, z2, ts)
92
93         imgs = model.decode(zs).clamp(0, 1) # (steps,3,32,32)
94         return imgs
95

```

```

1 ##### Cell 2.2 Run interpolation using your trained model (Conv-VAE)
2 # Choose which trained VAE to use:
3 model_to_use = conv_vae # <-- recommended for Task 2 (better spatial smoothness)
4
5
6 # Pick two CIFAR-10 images with different labels
7 (a_img, a_lbl, a_idx), (b_img, b_lbl, b_idx) = pick_two_images_different_classes(test_ds, seed=7)
8
9 print(f"Endpoint A: idx={a_idx}, class={CIFAR10_CLASSES[a_lbl]} ({a_lbl})")
10 print(f"Endpoint B: idx={b_idx}, class={CIFAR10_CLASSES[b_lbl]} ({b_lbl})")
11
12 # Show the original endpoints
13 show_grid(torch.stack([a_img, b_img]), title="Original endpoint images (from CIFAR-10)", nrow=2, figsize=(6,3))
14
15 # Interpolate in latent space and decode
16 STEPS = 12 # number of images in the interpolation grid (includes endpoints)
17 interp_imgs = interpolate_latent(
18     model=model_to_use.to(device),
19     x1=a_img,
20     x2=b_img,
21     steps=STEPS,
22     method="lerp",

```

```

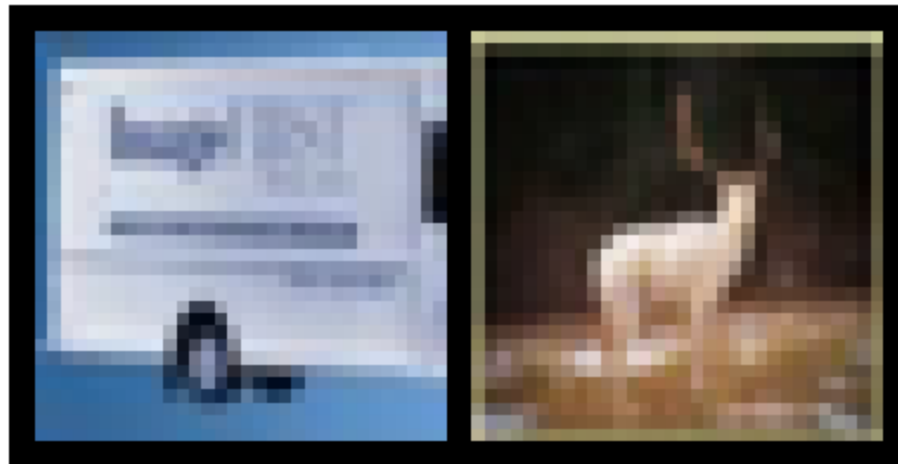
23     use_mu=True,      # deterministic, smoother
24     device=device
25 )
26
27 # Visualize interpolation result as a grid (left -> right transition)
28 show_grid(interp_imgs, title=f"Latent interpolation ({model_to_use.__class__.__name__}) - left→right transition", nrow=STEPS, figsize=(12, 4))
29

```

Endpoint A: idx=9449, class=truck (9)

Endpoint B: idx=6250, class=deer (4)

Original endpoint images (from CIFAR-10)



Latent interpolation (ConvVAE) — left→right transition



```

1 ##### Cell 2.3 Compare Conv-VAE vs FC-VAE interpolation side-by-side

```

```

2 STEPS = 12

```

```

3

```

```

4 interp_conv = interpolate_latent(conv_vae.to(device), a_img, b_img, steps=STEPS, method="lerp", use_mu=True, device=device)

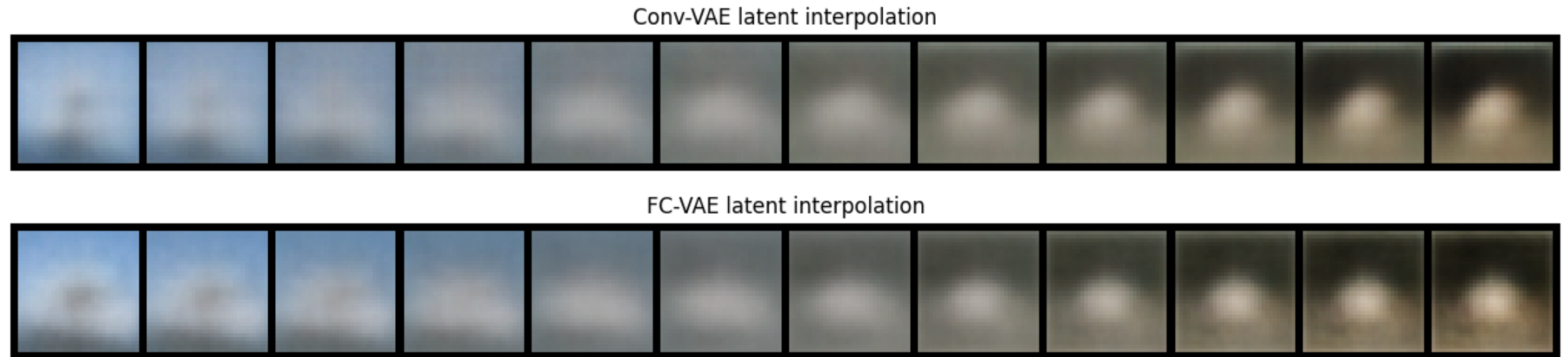
```

```

5 interp_fc   = interpolate_latent(fc_vae.to(device), a_img, b_img, steps=STEPS, method="lerp", use_mu=True, device=device)

```

```
6
7 show_grid(interp_conv, title="Conv-VAE latent interpolation", nrow=STEPS, figsize=(16,3))
8 show_grid(interp_fc, title="FC-VAE latent interpolation", nrow=STEPS, figsize=(16,3))
9
```



Note:

Interpolation is done in the latent space between two encoded points, producing intermediate latent vectors.

Decoding those vectors shows whether the learned latent space is smooth: adjacent frames should change gradually rather than jump.

Results interpretation

In sections B (2.2) and C (2.3), latent-space interpolation indicates that the Variational Autoencoder (VAE) learned a relatively smooth latent structure for CIFAR-10 images. As the latent representation transitions from the “truck” endpoint to

the “deer” endpoint, the decoded outputs change gradually without abrupt discontinuities. This suggests that nearby points in the latent space map to visually similar images, consistent with a smooth latent manifold.

Notably, the Conv-VAE produces more spatially consistent intermediate images during interpolation. Coarse shapes and background color regions evolve more smoothly and preserve a clearer overall layout, reflecting the convolutional encoder/decoder’s ability to retain local and hierarchical spatial features. In contrast, the FC-VAE interpolation appears more washed out and less structured, which is expected when images are flattened and spatial relationships are not modeled explicitly.

Using $\beta = 1.5$ increases the weight of the KL term, encouraging the latent distribution to remain closer to the standard normal prior. This typically promotes a smoother, more regular latent space, but it can also reduce sharpness and yield blurrier, more “average” outputs. In our results, the Conv-VAE still preserves more coherent structure than the FC-VAE under this stronger regularization, suggesting that convolutional inductive bias remains beneficial even with increased KL pressure.

Overall, while some blurriness remains, common for VAEs trained with pixel-wise reconstruction losses, the Conv-VAE interpolation demonstrates better preservation of spatial relationships and more coherent intermediate images than the fully connected VAE.

✓ Task3

Train the VAE on a new dataset of your choice (e.g., CelebA for faces), and visualize generated samples. Experiment with sampling from different regions of the latent space and analyze how the generated outputs vary based on different latent vectors.

References / Citations

- AntixK. (n.d.). PyTorch-VAE [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/AntixK/PyTorch-VAE>
- moshesipper. (n.d.). vae-torch-celeba [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/moshesipper/vae-torch-celeba>

- bhpfelix. (n.d.). Variational-Autoencoder-PyTorch [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/bhpfelix/Variational-Autoencoder-PyTorch>
- julian-8897. (n.d.). conv-vae-pytorch [GitHub repository]. GitHub. Retrieved January 31, 2026, from <https://github.com/julian-8897/conv-vae-pytorch>

```
1 from google.colab import drive
2 from pathlib import Path
3 drive.mount('/content/drive')
4
5 import os
6
7 ZIP_PATH = "/content/drive/MyDrive/DS552_CS552_Generative_Artificial_Intelligence/Assignment_3_Variational_Autoencoders_VAEs/data/i
8
9 print("ZIP_PATH:", ZIP_PATH)
10 print("Exists?", os.path.exists(ZIP_PATH))
11
12 if os.path.exists(ZIP_PATH):
13     print("File size (MB):", os.path.getsize(ZIP_PATH) / (1024**2))
14 else:
15     print("❌ File not found – path is wrong or Drive not mounted")
16
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

ZIP_PATH: /content/drive/MyDrive/DS552_CS552_Generative_Artificial_Intelligence/Assignment_3_Variational_Autoencoders_VAEs/data/img_align_celeba.zip

Exists? True

File size (MB): 1376.620138168335

```
1 ##### Cell 3.1 – Setup (imports, seed, device)
2 import zipfile
3 from glob import glob
4 from PIL import Image
5 from torch.utils.data import Dataset, random_split
6
7 # Unzip once (if not already extracted)
8 IMG_DIR = os.path.join("./data", "celeba_images", "img_align_celeba")
9 ZIP_PATH = "/content/drive/MyDrive/DS552_CS552_Generative_Artificial_Intelligence/Assignment_3_Variational_Autoencoders_VAEs/data/i
```



```

10
11 os.makedirs(os.path.dirname(IMG_DIR), exist_ok=True)
12 if not os.path.exists(IMG_DIR) or len(os.listdir(IMG_DIR)) == 0:
13     print("Extracting CelebA images...")
14     with zipfile.ZipFile(ZIP_PATH, "r") as zf:
15         zf.extractall(os.path.join("./data", "celeba_images"))
16 else:
17     print("Images already extracted:", IMG_DIR)
18
19 class FlatImageFolder(Dataset):
20     def __init__(self, img_dir, transform=None, max_items=None):
21         self.paths = sorted(glob(os.path.join(img_dir, "*.jpg")))
22         if max_items is not None:
23             self.paths = self.paths[:max_items]
24         self.transform = transform
25
26     def __len__(self):
27         return len(self.paths)
28
29     def __getitem__(self, idx):
30         img = Image.open(self.paths[idx]).convert("RGB")
31         if self.transform:
32             img = self.transform(img)
33         return img, 0 # dummy label
34
35 dataset = FlatImageFolder(IMG_DIR, transform=transform)
36
37 # Make your own split
38 val_frac = 0.1
39 n_val = int(len(dataset) * val_frac)
40 n_train = len(dataset) - n_val
41
42 train_ds, valid_ds = random_split(
43     dataset, [n_train, n_val],
44     generator=torch.Generator().manual_seed(42)
45 )
46
47 train_loader = DataLoader(train_ds, batch_size=BATCH_SIZE, shuffle=True, num_workers=0, pin_memory=(device.type=="cuda"))
48 valid_loader = DataLoader(valid_ds, batch_size=BATCH_SIZE, shuffle=False, num_workers=0, pin_memory=(device.type=="cuda"))

```

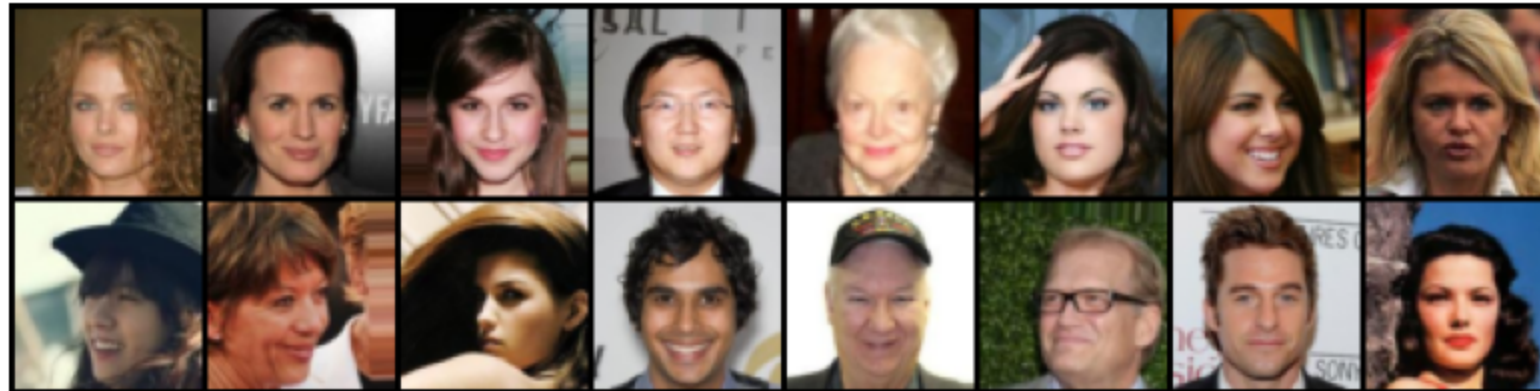
```

49
50 print("Train size:", len(train_ds), "Valid size:", len(valid_ds))
51
52 x0, _ = next(iter(train_loader))
53 show_grid(x0[:16], title="CelebA images (custom loader)", nrow=8, figsize=(10,5))
54

```

Images already extracted: ./data/celeba_images/img_align_celeba
Train size: 182340 Valid size: 20259

CelebA images (custom loader)



```

1 # Cell 3.2 – Load CelebA (faces) from local ZIP (no torchvision download)
2
3 import os, zipfile
4 from glob import glob
5 from PIL import Image
6 from torch.utils.data import Dataset, random_split, DataLoader
7 import torchvision.transforms as T
8
9 # --- Config ---
10 IMG_SIZE = 64
11 BATCH_SIZE = 128 # if OOM, try 64
12 NUM_WORKERS = 0 # recommended in Colab (avoids multiprocessing shutdown noise)
13
14 MAX_IMAGES = None # optional quick run: e.g., 50000 (uses only first N images after extraction)
15
16 transform = T.Compose([

```

```
17     T.CenterCrop(178),
18     T.Resize((IMG_SIZE, IMG_SIZE)),
19     T.ToTensor(),
20 ] )
21
22 # Your Drive zip path (works for you already)
23 ZIP_PATH = "/content/drive/MyDrive/DS552_CS552_Generative_Artificial_Intelligence/Assignment_3_Variational_Autoencoders_VAEs/data/j
24
25 # Extract destination
26 IMG_DIR = os.path.join("./data", "celeba_images", "img_align_celeba")
27 os.makedirs(os.path.dirname(IMG_DIR), exist_ok=True)
28
29 # Extract once
30 if not os.path.exists(IMG_DIR) or len(os.listdir(IMG_DIR)) == 0:
31     print("Extracting CelebA images...")
32     with zipfile.ZipFile(ZIP_PATH, "r") as zf:
33         zf.extractall(os.path.join("./data", "celeba_images"))
34 else:
35     print("Images already extracted:", IMG_DIR)
36
37 class FlatImageFolder(Dataset):
38     def __init__(self, img_dir, transform=None, max_items=None):
39         self.paths = sorted(glob(os.path.join(img_dir, "*.jpg")))
40         if max_items is not None:
41             self.paths = self.paths[:max_items]
42         self.transform = transform
43
44     def __len__(self):
45         return len(self.paths)
46
47     def __getitem__(self, idx):
48         img = Image.open(self.paths[idx]).convert("RGB")
49         if self.transform:
50             img = self.transform(img)
51         return img, 0 # dummy label (VAE doesn't need labels)
52
53 dataset = FlatImageFolder(IMG_DIR, transform=transform, max_items=MAX_IMAGES)
54
55 # Make your own split (90/10)
```

```
56 val_frac = 0.1
57 n_val = int(len(dataset) * val_frac)
58 n_train = len(dataset) - n_val
59
60 train_ds, valid_ds = random_split(
61     dataset, [n_train, n_val],
62     generator=torch.Generator().manual_seed(42)
63 )
64
65 train_loader = DataLoader(
66     train_ds, batch_size=BATCH_SIZE, shuffle=True,
67     num_workers=NUM_WORKERS, pin_memory=(device.type=="cuda")
68 )
69 valid_loader = DataLoader(
70     valid_ds, batch_size=BATCH_SIZE, shuffle=False,
71     num_workers=NUM_WORKERS, pin_memory=(device.type=="cuda")
72 )
73
74 print("Train size:", len(train_ds), "| Valid size:", len(valid_ds))
75
76 # Quick visual sanity-check
77 x0, _ = next(iter(train_loader))
78 show_grid(x0[:16], title="CelebA samples (custom loader, 64x64)", nrow=8, figsize=(10,5))
79
```

Images already extracted: ./data/celeba_images/img_align_celeba
Train size: 182340 | Valid size: 20259

CelebA samples (custom loader, 64x64)



```
1 ##### Cell 3.3 – Conv-VAE for 64x64
2 class ConvVAE64(nn.Module):
3     def __init__(self, latent_dim=128):
4         super().__init__()
5         self.latent_dim = latent_dim
6
7         # 64 -> 32 -> 16 -> 8 -> 4
8         self.encoder = nn.Sequential(
9             nn.Conv2d(3, 32, 4, 2, 1), # (B,32,32,32)
10            nn.BatchNorm2d(32),
11            nn.ReLU(inplace=True),
12
13            nn.Conv2d(32, 64, 4, 2, 1), # (B,64,16,16)
14            nn.BatchNorm2d(64),
15            nn.ReLU(inplace=True),
16
17            nn.Conv2d(64, 128, 4, 2, 1), # (B,128,8,8)
18            nn.BatchNorm2d(128),
19            nn.ReLU(inplace=True),
20
21            nn.Conv2d(128, 256, 4, 2, 1), # (B,256,4,4)
22            nn.BatchNorm2d(256),
```

```

23         nn.ReLU(inplace=True),
24     )
25
26     self.enc_out_dim = 256 * 4 * 4
27     self.fc_mu = nn.Linear(self.enc_out_dim, latent_dim)
28     self.fc_logvar = nn.Linear(self.enc_out_dim, latent_dim)
29
30     self.fc_dec = nn.Linear(latent_dim, self.enc_out_dim)
31
32     # 4 -> 8 -> 16 -> 32 -> 64
33     self.decoder = nn.Sequential(
34         nn.ConvTranspose2d(256, 128, 4, 2, 1), # (B,128,8,8)
35         nn.BatchNorm2d(128),
36         nn.ReLU(inplace=True),
37
38         nn.ConvTranspose2d(128, 64, 4, 2, 1), # (B,64,16,16)
39         nn.BatchNorm2d(64),
40         nn.ReLU(inplace=True),
41
42         nn.ConvTranspose2d(64, 32, 4, 2, 1), # (B,32,32,32)
43         nn.BatchNorm2d(32),
44         nn.ReLU(inplace=True),
45
46         nn.ConvTranspose2d(32, 3, 4, 2, 1), # (B,3,64,64)
47         nn.Sigmoid(), # output in [0,1]
48     )
49
50     def encode(self, x):
51         h = self.encoder(x).view(x.size(0), -1)
52         mu = self.fc_mu(h)
53         logvar = self.fc_logvar(h)
54         return mu, logvar
55
56     def reparameterize(self, mu, logvar):
57         std = torch.exp(0.5 * logvar)
58         eps = torch.randn_like(std)
59         return mu + eps * std
60
61     def decode(self, z):

```

```

62         h = self.fc_dec(z).view(z.size(0), 256, 4, 4)
63         return self.decoder(h)
64
65     def forward(self, x):
66         mu, logvar = self.encode(x)
67         z = self.reparameterize(mu, logvar)
68         recon = self.decode(z)
69         return recon, mu, logvar
70

```

```

1 ##### Cell 3.4 – Loss + Training Loop
2 def kl_divergence(mu, logvar):
3     return -0.5 * torch.sum(1 + logvar - mu.pow(2) - logvar.exp())
4
5 def recon_loss_fn(recon_x, x, kind="mse"):
6     # sum over batch+pixels
7     if kind == "bce":
8         return F.binary_cross_entropy(recon_x, x, reduction="sum")
9     return F.mse_loss(recon_x, x, reduction="sum")
10
11 def vae_loss(recon_x, x, mu, logvar, beta=1.0, recon_kind="mse"):
12     recon = recon_loss_fn(recon_x, x, kind=recon_kind)
13     kl = kl_divergence(mu, logvar)
14     total = recon + beta * kl
15     bs = x.size(0)
16     return total / bs, recon / bs, kl / bs
17
18 def train_vae(model, train_loader, valid_loader, epochs=10, lr=2e-4, beta=1.0, recon_kind="mse"):
19     model = model.to(device)
20     opt = torch.optim.Adam(model.parameters(), lr=lr)
21
22     use_amp = (device.type == "cuda")
23     scaler = torch.cuda.amp.GradScaler(enabled=use_amp)
24
25     history = {"train_total": [], "train_recon": [], "train_kl": [],
26              "valid_total": [], "valid_recon": [], "valid_kl": []}
27
28     for epoch in range(1, epochs + 1):

```

```

29     # ---- train ----
30     model.train()
31     t_total = t_recon = t_kl = 0.0
32     n_train = 0
33
34     pbar = tqdm(train_loader, desc=f"Epoch {epoch}/{epochs} [train]", leave=False)
35     for x, _ in pbar:
36         x = x.to(device, non_blocking=True)
37         opt.zero_grad(set_to_none=True)
38
39         with torch.cuda.amp.autocast(enabled=use_amp):
40             recon, mu, logvar = model(x)
41             loss, rloss, kl = vae_loss(recon, x, mu, logvar, beta=beta, recon_kind=recon_kind)
42
43             scaler.scale(loss).backward()
44             scaler.step(opt)
45             scaler.update()
46
47             bs = x.size(0)
48             t_total += loss.item() * bs
49             t_recon += rloss.item() * bs
50             t_kl += kl.item() * bs
51             n_train += bs
52             pbar.set_postfix({"loss": loss.item(), "recon": rloss.item(), "kl": kl.item()})
53
54     # ---- valid ----
55     model.eval()
56     v_total = v_recon = v_kl = 0.0
57     n_valid = 0
58     with torch.no_grad():
59         for x, _ in tqdm(valid_loader, desc=f"Epoch {epoch}/{epochs} [valid]", leave=False):
60             x = x.to(device, non_blocking=True)
61             recon, mu, logvar = model(x)
62             loss, rloss, kl = vae_loss(recon, x, mu, logvar, beta=beta, recon_kind=recon_kind)
63
64             bs = x.size(0)
65             v_total += loss.item() * bs
66             v_recon += rloss.item() * bs
67             v_kl += kl.item() * bs

```



```

68         n_valid += bs
69
70     train_total = t_total / n_train
71     train_recon = t_recon / n_train
72     train_kl = t_kl / n_train
73
74     valid_total = v_total / n_valid
75     valid_recon = v_recon / n_valid
76     valid_kl = v_kl / n_valid
77
78     history["train_total"].append(train_total)
79     history["train_recon"].append(train_recon)
80     history["train_kl"].append(train_kl)
81
82     history["valid_total"].append(valid_total)
83     history["valid_recon"].append(valid_recon)
84     history["valid_kl"].append(valid_kl)
85
86     print(
87         f"Epoch {epoch:02d}/{epochs} | "
88         f"Train total={train_total:.4f} recon={train_recon:.4f} kl={train_kl:.4f} | "
89         f"Valid total={valid_total:.4f} recon={valid_recon:.4f} kl={valid_kl:.4f}"
90     )
91
92     return history
93

```

```

1 ##### Cell 3.5 – Train on CelebA + quick reconstructions
2
3
4 LATENT_DIM = 128
5 EPOCHS = 10      # 10 for this assignment
6 LR = 2e-4
7 BETA = 1.0
8 RECON = "mse"
9
10 celeba_vae = ConvVAE64(latent_dim=LATENT_DIM).to(device)
11

```

```
12 hist_celeba = train_vae(
13     celeba_vae, train_loader, valid_loader,
14     epochs=EPOCHS, lr=LR, beta=BETA, recon_kind=RECON
15 )
16
17 # (Sanity check) Save weights so we don't lose training when Colab resets
18 torch.save(
19     {"state_dict": celeba_vae.state_dict(), "latent_dim": LATENT_DIM},
20     "celeba_convvae64.pt"
21 )
22
23 # Reconstructions on a validation batch (use inference_mode for speed)
24 celeba_vae.eval()
25 xv, _ = next(iter(valid_loader))
26 xv = xv.to(device)[:8]
27
28 with torch.inference_mode():
29     recon, _, _ = celeba_vae(xv)
30
31 viz = torch.cat([xv, recon.clamp(0, 1)], dim=0)
32 show_grid(viz, title="CelebA reconstructions: original (top) vs recon (bottom)", nrow=8, figsize=(14,4))
33
34
```


/tmp/ipython-input-3623721387.py:23: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler`

```
1 ##### Cell 3.6 – Generate samples from the prior
2 @torch.no_grad()
3 def sample_prior(model, latent_dim, n=16, scale=1.0):
4     model.eval()
5     z = torch.randn(n, latent_dim, device=device) * scale
6     x = model.decode(z).clamp(0, 1)
7     return x
8
9 samples = sample_prior(celeba_vae.to(device), LATENT_DIM, n=32, scale=1.0)
10 show_grid(samples, title="CelebA Conv-VAE samples (z ~ N(0,1))", nrow=8, figsize=(12,6))
11
12
```

Epoch 03/10 | Train total=194.3585 recon=132.3120 kl=62.0465 | Valid total=191.6297 recon=129.3513 kl=62.2785

Epoch 04/10 | Train total=190.4936 recon=128.2034 kl=62.2902 | Valid total=187.7902 recon=124.5183 kl=63.2719

Epoch 05/10 | Train total=187.9731 recon=125.5713 kl=62.4018 | Valid total=187.2936 recon=124.4094 kl=62.8842

Epoch 06/10 | Train total=186.1303 recon=123.6210 kl=62.5094 | Valid total=184.6971 recon=122.6068 kl=62.0903

Epoch 07/10 | Train total=184.6992 recon=122.1345 kl=62.5647 | Valid total=184.0491 recon=121.2157 kl=62.8334

Epoch 08/10 | Train total=183.5224 recon=120.9248 kl=62.5975 | Valid total=182.9543 recon=119.8686 kl=63.0857

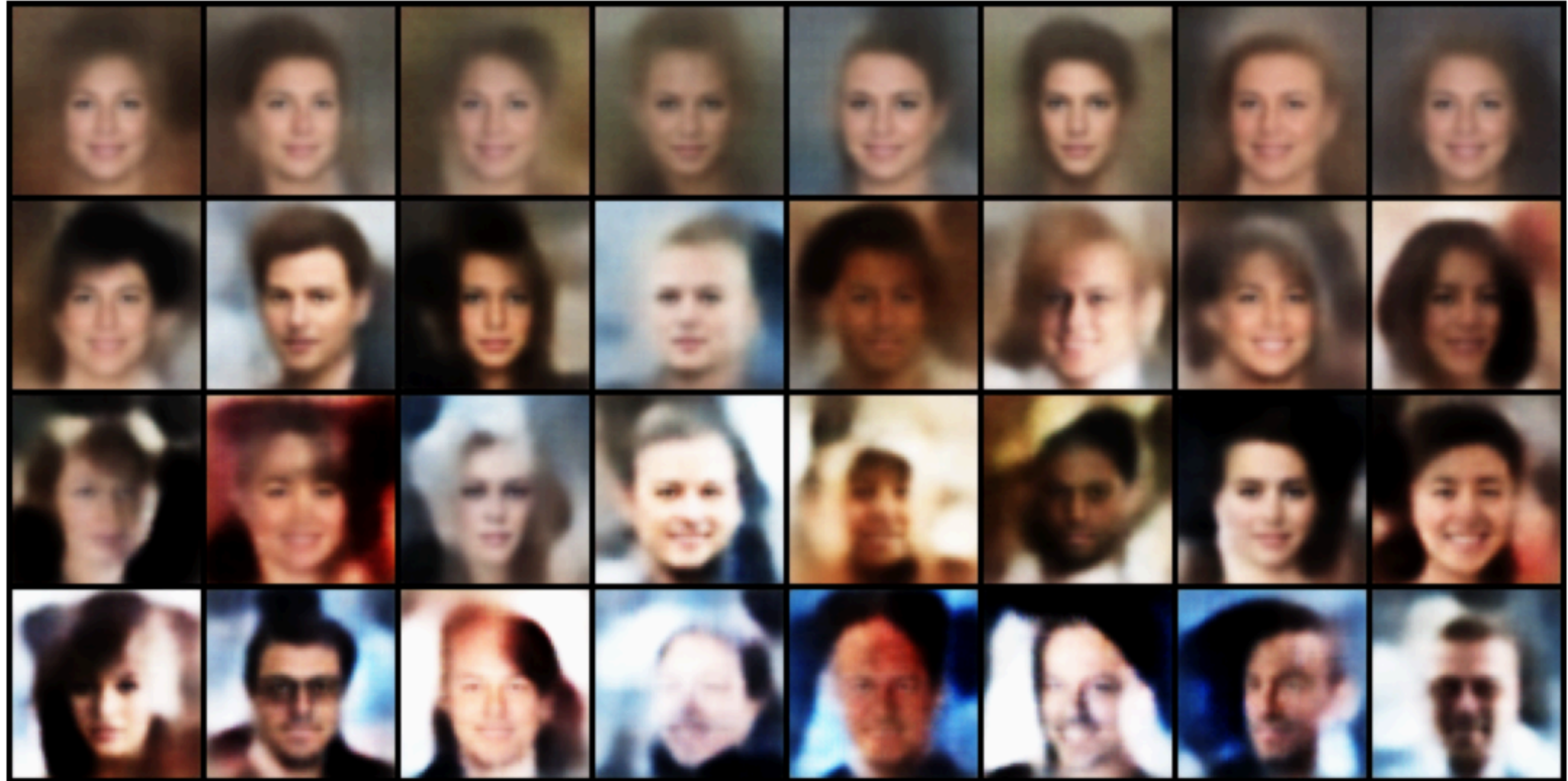
Epoch 09/10 | Train total=182.4185 recon=119.8807 kl=62.6147 | Valid total=182.4851 recon=119.4100 kl=63.4100

CelebA Conv-VAE samples ($z \sim N(0,1)$)

```
1 ##### Cell 3.7 – Explore “different regions” of the latent space
2 #“Temperature” / scale sampling (near vs far from origin)
3 scales = [0.5, 1.0, 1.5, 2.0]
4 rows = []
5 for s in scales:
6     rows.append(sample_prior(celeba_vae.to(device), LATENT_DIM, n=8, scale=s))
7 grid = torch.cat(rows, dim=0)
8
9 show_grid(
10     grid,
```

```
11 title="Latent sampling scale: each row uses  $z \sim N(0, \text{scale}^2 I)$  (top=0.5  $\rightarrow$  bottom=2.0)",
12 nrow=8,
13 figsize=(14,7)
14 )
15
```

Latent sampling scale: each row uses $z \sim N(0, \text{scale}^2 I)$ (top=0.5 $\rightarrow$ bottom=2.0)



Cell 3.7 Analysis latent "temperatur" / scale sampling

Each row samples $z \sim N(0, \text{scale}^2 I)$, so larger scale moves latent points farther from the origin. At scale = 0.5, faces look more consistent and "average," reflecting sampling from high-density latent regions. At scale = 1.0–1.5, outputs become more diverse (hair, shape, lighting) while still face-like. At scale = 2.0, images show more blur and artifacts, suggesting off-manifold sampling. Overall, smaller scales give higher quality but less variety, while larger scales increase diversity at the cost of realism.

```
1 ##### Cell 3.8 – Posterior neighborhood sampling (around a real image)
2 @torch.no_grad()
3 def posterior_neighborhood(model, x, alphas=(0.0, 0.25, 0.5, 1.0), n_per_alpha=6):
4     model.eval()
5     x = x.unsqueeze(0).to(device) # (1,3,64,64)
```